Airborne Lead and Birth Outcomes: Evidence from Leaded Fuel in Auto Racing

Yuulin An

Department of Economics, Colorado State University, Fort Collins, CO

October 30, 2026

JOB MARKET PAPER

LATEST VERSION

Abstract

The National Association for Stock Car Auto Racing (NASCAR) continued to use leaded gasoline until 2007, despite the removal of leaded gasoline from on-road vehicles long ago. Exploiting this natural experiment, I study how airborne lead exposure affected birth outcomes using administrative birth records linked to race events at Texas Motor Speedway (TMS) in Fort Worth, Texas. Before the transition to unleaded fuel, race exposure among mothers living within 25 km of TMS is associated with lower infant birthweight, shorter gestation, and a higher probability of low birthweight while the corresponding birthweight and gestational-age estimates are near zero after the transition. To account for how lead accumulates in the body, I develop a pregnancy-specific physiologically based pharmacokinetic (PBPK) model that simulates maternal daily blood lead trajectories. I find that each additional race during pregnancy lowers birthweight by about 14 grams, and infants of mothers at the 75th percentile of simulated cumulative blood lead weigh about 38 grams less than those of mothers at the 25th percentile. Suggestive evidence points to gestational hypertension as a mechanism through which lead shortens gestation. These findings suggest that prenatal exposure to airborne lead can affect maternal and infant health.

JEL classifications: Q53, I18, J13, Q58

Keywords: Lead exposure, birthweight, prenatal health, NASCAR, leaded gasoline, PBPK model

1 Introduction

Lead (Pb) is one of the most well-established environmental neurotoxins and has been associated with numerous adverse outcomes including cognitive impairment, increased criminal behavior, and higher mortality (Talayero et al., 2023; Lanphear et al., 2019; Deryugina et al., 2019; Knittel et al., 2016; Stieb et al., 2012; Currie and Schmieder, 2008). Growing evidence on the harmful effects of lead prompted many countries to adopt policies aimed at reducing exposure beginning in the 1970s. One of the most consequential interventions was the phaseout of leaded gasoline from passenger vehicles which produced substantial public health gains. Despite this, lead is still present today in sources such as fuel used in piston-engine aircraft, paint in older housing, and drinking water. Notably, the National Association for Stock Car Auto Racing (NASCAR), the largest racing organization in the United States, continued to use leaded racing fuel until late 2006 to early 2007 before switching to unleaded gasoline. I exploit this transition as a quasi-experimental setting to estimate the effect of lead exposure on birth outcomes among mothers living near Texas Motor Speedway (TMS) in Fort Worth, Texas between 2003 and 2010. Before the transition, races generated concentrated bursts of airborne lead emissions surrounding the racetrack. After the transition, racing continued with unleaded fuel which provides a clean comparison to isolate the effect of lead exposure from other race-related activities.

I identify the effect of prenatal lead exposure entirely from within-TMS variation as the traditional cross-track comparison approach may produce unreliable estimates in this setting. Furthermore, differences across racetracks and the sociodemographic characteristics of surrounding communities make it difficult to construct a credible comparison group. Hence, the within-TMS design exploits four sources of variation in exposure among mothers living near the same racetrack. First, mothers living closer to TMS are likely to receive greater lead exposure as lead concentrations decline with distance from the track. Second, wind direction on race days is plausibly exogenous to maternal characteristics and residential sorting so mothers residing downwind of the racetrack are more exposed than mothers outside the lead plume. Third, the gestational timing of each race depends on the mother's conception date. Mothers who live the same distance from TMS but conceive just a few weeks apart may experience different patterns of race exposure across trimesters. Finally, emissions vary across races with caution flags, accidents, and race length which generates variation in the amount of lead released that is plausibly unrelated to maternal health.

To estimate the effect of lead exposure on birth outcomes, I construct five different exposure measures and estimate split-sample regressions comparing births before and after

the transition from leaded to unleaded racing fuel. This approach follows a difference-in-differences logic since identification rests on the assumption that factors correlated with race-day intensity such as traffic, noise, and pollution operate similarly across the leaded and unleaded periods. A negative and statistically significant relationship between lead exposure and birth outcomes in the leaded period and a null result in the unleaded period would be consistent with lead as the mechanism driving the effect. Although the effect of lead can be estimated using a simple reduced-form estimation, this approach does not account for how lead accumulates within a body, particularly during pregnancy. Lead can accumulate in maternal blood, tissues, and bone and be remobilized from bone back to blood in later trimesters. To account for these dynamics, I developed a pregnancy-specific Physiologically-Based Pharmacokinetic (PBPK) model that translates each mother's discrete race-day exposure pulses into daily blood lead trajectories over a 280-day pregnancy. The model compartmentalizes lead in the maternal blood, tissues, and bones and returns trimester-specific blood lead means, cumulative blood lead over pregnancy (AUC), and maximum blood lead while capturing the remobilization of lead stored in maternal bones during her later trimesters.

Before the transition, each additional race during pregnancy lowers birthweight by 14.5 grams, or 0.4 percent of mean birthweight, and a doubling of the lead emitted during pregnancy lowers birthweight by about 27 grams and shortens gestation by about one day. After the transition, when the same races continued without lead, the corresponding estimates are small and statistically insignificant. Comparing the two periods, the effect of lead emissions over the full pregnancy is larger before the transition than after it. The difference is statistically significant at the 1 percent level for gestational age, but only at the 10 percent level for birthweight. The shorter gestation reflects a higher risk of preterm birth. By trimester, races during the second trimester have the largest effect. Before the transition, the effects come almost entirely from infants born in 2005 and 2006. Several tests in the Appendix including the second race weekend that TMS added in 2005 do not explain this pattern. Birthweight also rises with distance from the track before the transition, by 1.67 grams per kilometer, while this gradient disappears afterward. Holding the number of races fixed, mothers who were downwind of TMS on every race day during pregnancy delivered infants about 40 grams lighter than mothers who were never downwind, although the post-transition wind estimates are positive rather than zero. The PBPK model yields a similar pattern. Moving from the 25th to the 75th percentile of simulated cumulative blood lead over the pregnancy lowers birthweight by about 38 grams. The birthweight and gestational age results are consistent with a pathway through shorter gestation, and suggestive evidence points to gestational hypertension as a channel.

A large body of work documents that exposure to ambient pollution reduces birthweight, increases the risk of preterm birth, and worsens a wide range of other infant health outcomes (Pedersen et al., 2013; Currie, 2013; Stieb et al., 2012; Currie and Neidell, 2005). A related literature also found that lead reduces educational attainment, lowers adult earnings, and generates persistent socioeconomic disadvantages over the life cycle (Talayero et al., 2023; Jenkins et al., 2022; Boyle et al., 2021; Aizer et al., 2018; Liu et al., 2014). Yet, less is known about the effects of prenatal exposure to localized sources of airborne lead and how these effects vary with the timing of exposure during pregnancy despite the fact that lead readily crosses the placenta and can affect fetal development in utero (Santana et al., 2023; Yüksel et al., 2022; Zhu et al., 2010). This paper contributes to these literatures in three ways. First, I provide causal evidence on the effects of a localized source of airborne lead on maternal and infant health using within-track variation in exposure. I compare mothers living near the same racetrack and exploit variation in race timing, emissions, distance, and wind direction rather than relying on comparisons across racetracks, which produced unreliable estimates in this setting. Second, I integrate a pregnancy-specific PBPK model to account for how lead accumulates and remobilizes throughout pregnancy. The model translates discrete exposures on race days into daily maternal blood lead levels and allows exposure in one trimester to affect blood lead levels in later trimesters. Third, I measure exposure at the census-tract level and incorporate wind direction on race days to separate mothers residing downwind of the racetrack and those outside the lead plume rather than relying on solely on distance.

The remainder of the paper proceeds as follows. Section 2 provides background on leaded gasoline and NASCAR's switch to unleaded gasoline in 2007. Section 3 describes the birth records, race data, and weather data. Section 4 describes how I construct race-level lead exposure measures along with the PBPK model used to trace lead exposure over the course of pregnancy. Section 5 describes the empirical strategy, which uses variation in the timing of exposure, distance from TMS, and wind direction to measure exposure by trimester and over the full pregnancy. Section 6 presents the main results and examines heterogeneity by distance and wind direction, the PBPK estimates, and maternal hypertension as a potential mechanism. The section also briefly compares the results with a monitor-based exposure measure and with other Texas racetracks. Section 7 concludes. Finally, the Appendix discusses robustness and falsification tests and provides a monitor-based exposure measure as an alternative specification.

2 Background and Institutional Setting

Leaded gasoline remained common in motorsports long after it disappeared from commercial passenger vehicles. Lead served as an octane booster, which allows engines to operate at very high compression ratios without engine knock. Prior to the transition, NASCAR races used fuel containing approximately 4 to 5.2 grams of tetraethyl lead per gallon (Hollingsworth et al., 2025). The organization operates hundreds of races each year through both national and regional racing series. Its three national series are the Cup Series, Xfinity Series (currently called O'Reilly Auto Parts Series), and Camping World Truck Series (currently called Craftsman Truck Series) while its regional series includes the K&N Pro Series West (KNPSW). These series follow a strict guideline on the type of vehicles and fuel used in each race. NASCAR mandated all cars in a series to have the same standardized fuel supplied by its fuel sponsor, Sunoco (Hollingsworth and Rudik, 2021). Any variation in lead exposure would have been due to when and where races were held, as opposed to what fuel drivers or teams used.

Figure 1 shows TMS in 2023 in Fort Worth, Texas.[1] It is the primary venue in this study.[2] The 1.5-mile quad-oval racetrack occupies approximately 2.3 square miles and held its first major race in 1997 and has since become a top NASCAR active national venue. During the sample period (2003–2010), TMS hosted an average of six NASCAR races annually, typically in the spring and fall. Table 25 provides a full list of NASCAR race schedule at TMS between 2003 and 2010.

[1] Due to copyright restrictions, I cannot reproduce an image of TMS from 1998. However, the image can be viewed at: https://www.autoweek.com/racing/nascar/a41341864/nascar-playoff-race-chapter-texas-motor-speedway/.

[2] There are three other racetracks in Texas that hosted NASCAR races. I discussed them in Appendix J and why cross-track comparison produces unreliable estimates in this setting.

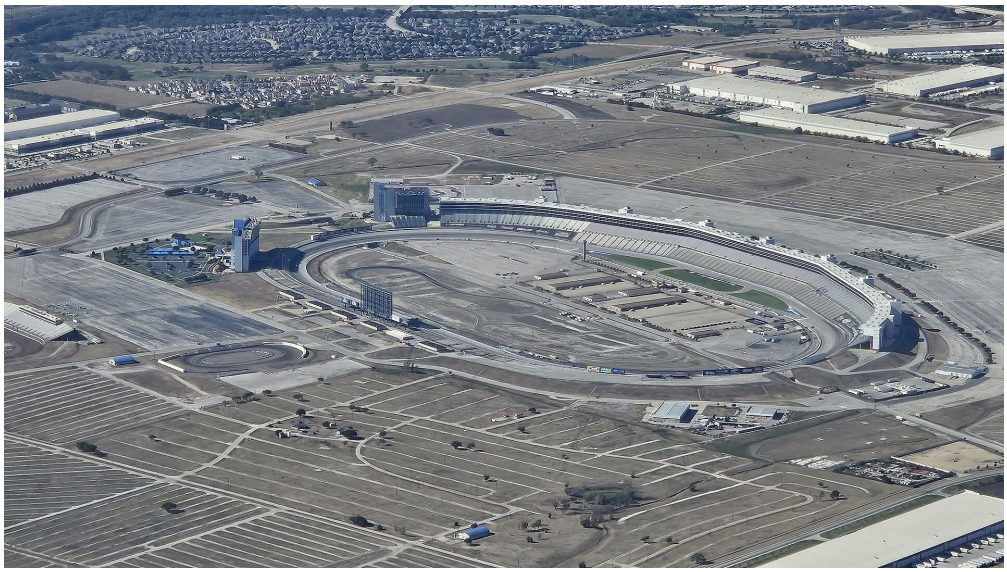


Figure 1. Texas Motor Speedway in 2023 via *Wikipedia*

The Clean Air Act of 1970 established the federal framework for regulating air pollution in the United States. Beginning in the 1970s, the EPA introduced regulations to gradually phase lead out of gasoline, partly due to growing concerns about the health effects of lead exposure. By 1996, leaded gasoline was completely phased out from on-road vehicles in the United States. Racing vehicles, however, were exempt from these requirements and continued to use leaded gasoline. NASCAR began transitioning to unleaded gasoline in 2006 and fully switched their national racing series to unleaded fuel in 2007. The Xfinity and Camping World Truck Series began using unleaded fuel on September 23, 2006, while the Cup Series continued using leaded gasoline through the end of the 2006 season and permanently switched to unleaded fuel on February 25, 2007.[3] In this setting, TMS's last leaded race was on November 5, 2006 while its first unleaded race was on April 14, 2007. Since TMS had no races in between these two dates, any post-transition cutoff date after February 25, 2007 is nearly four months after the last leaded race at TMS, so the exact choice of cutoff does not affect any mother's exposure. Hence, I follow the national switch and set the cutoff date to March 1, 2007. By then, roughly three half-lives of blood lead had passed since the last leaded race as lead in blood has a half-life of about 30 to 40 days. Lead stored in bone, however, can remain for years.

Given this staggered transition, I divide the sample into three windows: pre-transition, transitional, and post-transition. The pre-transition window includes births conceived before December 17, 2005 so that the third trimester ended before the Xfinity and Camping

[3]ARCA, another racing organization in the US that got acquired by NASCAR in 2018, also switched to unleaded fuel beginning in 2007. Throughout the sample, ARCA had only one race at TMS in 2010, and hence, the entire analysis rests on NASCAR races which I will mainly refer to throughout this paper.

World Truck Series began using unleaded fuel on September 23, 2006. The post-transition window includes births conceived on or after March 1, 2007. The transitional window includes births conceived between December 17, 2005 and February 28, 2007, and is excluded from the cross-sectional sample to prevent gestational overlap between the pre- and post-transition periods.

3 Data

I combine three primary data sources which are linked at the individual birth level.

Birth records. I use confidential Texas birth records from 2003 to 2010, obtained from the Texas Department of State Health Services (DSHS) Vital Statistics. These records contain information on exact birth timing and location, birth outcomes, maternal characteristics, pregnancy complications, and the census tract of maternal residence. The primary outcome is birthweight in grams. Secondary outcomes include gestational age in weeks, low birthweight status ($<$ 2,500 grams), and birthweight conditional on gestational age. I also look at gestational hypertension, eclampsia, and the five-minute Apgar score to examine potential mechanisms. Gestational hypertension is high blood pressure that develops after the 20th week of pregnancy while eclampsia is characterized by the onset of seizures associated with preeclampsia. The five-minute Apgar score assesses a newborn's condition five minutes after birth on a scale of 0 to 10 based on heart rate, skin color, reflex response, muscle tone, and breathing rate. Scores below 7 indicate that the newborn may need medical attention.

In addition, the 2008-2010 birth records do not report the exact date of birth. For these years, I impute the birth date using the mother's reported last menstrual period (LMP) and the LMP-based gestational length. I also use the LMP to estimate conception timing and assign race-day exposure to each trimester. This approach introduces potential measurement errors and attenuates the estimated effects toward zero as exposure may be misclassified due to the mother's reported LMP may be different from the actual date by several days. However, I use the reported LMP to estimate conception dates for all births, regardless of year. Since this measurement error is present both before and after the fuel transition, it is unlikely to explain why I find no effect after the transition. I also harmonize variables across years and restrict the sample to singleton births with an obstetric estimate of gestational age between 20 and 44 completed weeks.

For each birth, I calculate the start of pregnancy as

$$C_i = DOB_i - 7 \times G_i$$

where DOB_i is the date of birth and G_i is the LMP-based gestational length in weeks. I refer to C_i as the conception date throughout the analysis. Then, I assign every race in the sample period to a trimester relative to that pregnancy such that Trimester 1 (T1) covers conception through gestational day 97; T2 covers days 98-195; and T3 covers days 196-280, following ACOG conventions (ACOG, 2024). Each mother is matched to the nearest of the four Texas tracks, and the analysis sample is restricted to mothers whose nearest track is TMS and who reside within 25 km of it. Figure 2 shows a close up picture of all live births within 25 km of TMS. I construct exposure at the birth-by-race-day level and determine whether a race occurred during pregnancy, which trimester it fell in, how much lead was emitted, and whether the mother lived downwind on that day. Then I aggregate these to the birth level so each birth is one observation in the analysis.

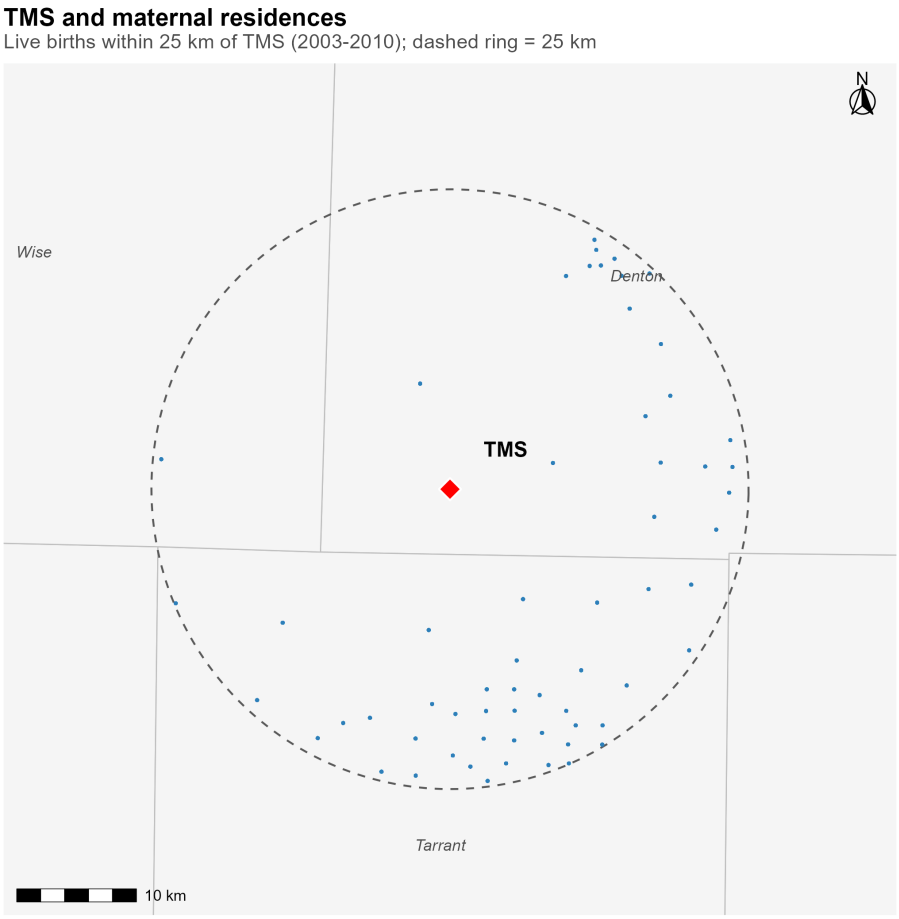


Figure 2. Red diamond indicates the TMS racetrack in Fort Worth while the blue circles indicate maternal residential locations, measured using census tract centroids, in the birth sample. The full sample size within 25 km of TMS between 2003-2010 is $N = 72,471$.

Race data. Race data come from Racing Reference which contains every NASCAR event held at TMS (Racing-Reference, nd). These data include the exact race date, track location, racing series, total laps, lap length, miles traveled, number of drivers, total race duration

in seconds, and other race characteristics. The data distinguish across NASCAR's major racing series including the Cup Series, Xfinity Series, and Truck Series, which is important because different series transitioned away from leaded fuel at different times.

Weather data. Weather data come from the National Oceanic and Atmospheric Administration's (NOAA) NCEP-NCAR Reanalysis 1 model (Kalnay et al., 2018). Wind direction, wind speed, precipitation, air temperature, and relative humidity on each race day are daily means at the reanalysis grid point nearest TMS. On a given race day, the wind therefore blows in the same direction across the study area, and whether a mother is downwind depends on where she lives relative to the track. For each birth $\times$ race day, I compute the bearing from the racetrack to the maternal census tract centroid. A mother is classified as "downwind" on race day j if the angle between the track-to-residence bearing and the wind direction on race days is within $45°$.

4 Exposure Construction

I construct several measures of race-related lead exposure during pregnancy. The two primary measures are based on the amount of lead emitted during races and the number of races that occur over the full pregnancy and in each trimester of pregnancy. The secondary measures are distance from the racetrack and wind direction on race days. Finally, I use a pregnancy-specific physiologically based pharmacokinetic (PBPK) model to account for the dynamics of lead within the maternal body.

4.1 Emissions and Race Counts

For each birth-race pair, I compute the lead emitted during the race and then sum it over the races that fall in each trimester and over the full pregnancy. Lead emitted at race j is calculated as:

$$\text{LeadEmitted}_j = \frac{\text{MilesTraveled}_j}{\text{MPG}} \times \text{LeadPerGallon} \tag{1}$$

where LeadEmitted_j is the total amount of lead in grams released during race j; MilesTraveled_j is the total miles driven by all cars during race j; MPG is miles per gallon and is assumed to equal five which is a typical fuel economy for NASCAR; and LeadPerGallon is set to 5.2 grams, the average pre-transition tetraethyl lead concentration per gallon (Hollingsworth et al., 2025). Before the transition, Equation 1 measures the lead actually released during a race. After the transition, races used unleaded fuel

and released no lead, so I apply the same formula to construct a counterfactual measure of the lead each race would have emitted had it still used leaded fuel.

For each mother i and trimester $k \in \{1, 2, 3\}$, I sum the total amount of lead emitted from Equation 1 across all races at TMS whose date falls within trimester k of her pregnancy:

$$E_{ik} = \sum_{j \in \text{trimester}k(i)} \text{LeadEmitted}_j. \tag{2}$$

Because the distribution is right-skewed, I take the log plus one transformation,

$$\widetilde{E}_{ik} = \log(1 + E_{ik}). \tag{3}$$

where $\widetilde{E}_{ik}$ is the log emission for mother i in trimester k. Summing over the three trimesters gives the amount of lead emitted over the full pregnancy, $E_i = \sum_{k=1}^{3} E_{ik}$, and its log, $\widetilde{E}_i = \log(1 + E_i)$. Every mother in the sample is exposed to at least one race during pregnancy whereas trimester k can have mothers with no race so that E_{ik} equals zero. Appendix C shows that the full-pregnancy estimates are similar when I use the logarithm of E_i without adding one, the inverse hyperbolic sine, emissions in kilograms, or emissions in levels. However, the trimester estimates depend on the units of emissions because many trimesters have no race, which is another reason I use race counts for the trimester results.

The second exposure measure is the number of races at TMS whose date falls within trimester k of mother i's pregnancy, R_{ik}, and over the full pregnancy, R_i. It treats every race as the same exposure and is defined identically before and after the transition, regardless of the fuel used.

Under the hypothesis that the pre-transition effect operates through lead, the post-transition coefficients on $\widetilde{E}_{ik}$, $\widetilde{E}_i$, R_{ik}, and R_i should be near zero. An effect after the transition, however, would mean that non-lead factors such as noise, traffic, and residential sorting near the track are driving the effect. In that case, it would be unwarranted to interpret the pre-transition effect as due to lead.

4.2 Distance and Wind Direction

Distance and wind direction provide two sources of variation in exposure in this study. First, a mother who lives closer to the racetrack should experience greater exposure to lead than a mother who lives farther away. Therefore, the effect of lead exposure on

birthweight should attenuate with distance from the racetrack. For each mother i, I compute distance, d_i, in kilometers from her census tract centroid to TMS and restrict the sample to mothers living within 25 km. Second, I classify mother i as living "downwind" of TMS on each race day if the angle between the bearing from TMS to her census tract centroid and the wind direction on race days at TMS is within $\pm 45°$. Figure 3 illustrates the downwind classification of mother i. If she resides within the $\pm 45°$ lead plume, she is considered residing downwind as depicted by Mother A while Mother B resides upwind outside the plume.

For each mother i, I construct two downwind measures. The first measure is the fraction of race days during pregnancy on which she was downwind of TMS,

$$\text{ShareDownwind}_i = \frac{\#\{\text{race days during pregnancy with } i \text{ downwind}\}}{\#\{\text{race days during pregnancy}\}}, \tag{4}$$

which equals one for mothers who reside downwind on every race day during pregnancy. If mother i was downwind of the racetrack on two of three race days during her pregnancy, for instance, then her downwind exposure share would be 67 percent. The second is a binary measure that captures whether a mother was downwind of TMS on at least one race day during pregnancy:

$$\text{AnyDownwind}_i = \mathbf{1}\big[i \text{ is downwind on at least one race day during pregnancy}\big]. \tag{5}$$

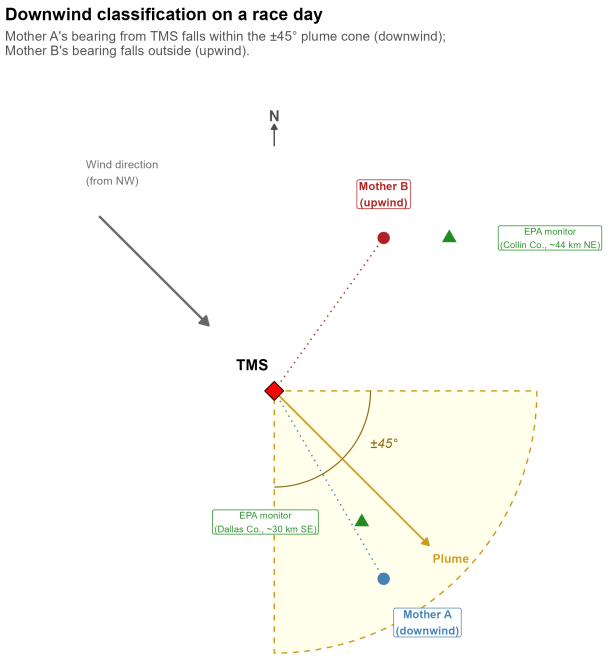


Figure 3. On each race day, mother i is classified as residing "downwind" of TMS if the angle between (a) the bearing from TMS to her census-tract centroid and (b) the wind direction on race days at TMS that day falls within $\pm 45°$.

4.3 Pregnancy-Specific PBPK Model

A central challenge in this setting is that lead exposure occurs intermittently. Lead has a half-life of 30 to 40 days in blood, weeks in soft tissue, and years to decades in bone. Thus, exposure occurring late in one trimester can increase maternal and fetal blood lead in subsequent trimesters. Moreover, lead stored in maternal bone can also slowly remobilize into the blood during the later part of pregnancy (Téllez-Rojo et al., 2004). Hence, the reduced-form specification (Equations 15 and 16) cannot capture these intra-pregnancy dynamics. It only identifies whether a mother was exposed and in which trimester, but it does not capture how much lead would remain and persist in her body across trimesters.

To account for this, I develop a pregnancy-specific physiologically based pharmacokinetic (PBPK) model to capture the accumulation and redistribution of lead across gestation. The model follows the general PBPK framework in O'Flaherty (1991); Oflaherty (1993, 1995) but simplifies its full physiological structure and underlying lead kinetics into three maternal compartments: blood, soft tissue, and bone. When mother i is exposed to lead released during a race, the dose enters her blood as a discrete pulse on the gestational day on which the race occurs. I define this dose using the race's lead emissions from Equation 1, expressed relative to the average leaded race occurring during pregnancy in the sample:

$$q_{ij} = \frac{\text{LeadEmitted}_j \times \text{Leaded}_j}{\overline{\text{LeadEmitted}}}, \qquad \overline{\text{LeadEmitted}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \text{LeadEmitted}_j \tag{6}$$

where LeadEmitted_j is the amount of lead, in grams, released during race j from Equation 1; Leaded_j equals one if race j used leaded fuel and zero otherwise; and $\mathcal{P}$ is the set of all mother-race pairs in which a leaded race occurs during the mother's pregnancy. $\overline{\text{LeadEmitted}}$ is therefore the average emissions of a leaded race in the full sample, weighted by the number of pregnancies each race overlaps. This makes q_{ij} a normalized dose relative to the average leaded race. A value of $q_{ij} = 2$, for example, means that race j emitted twice as much lead as the average leaded race while races using unleaded fuel have a dose of zero. Since the dose is based only on race-level emissions, all mothers exposed to the same race receive the same dose in the PBPK model. Race-day exposure is then represented as

$$E_i(t) = \sum_{j=1}^{J_i} q_{ij}\delta(t - t_{ij}) \tag{7}$$

where $E_i(t)$ is the total lead input from race-day exposure on gestational day t; J_i is the number of races during mother i's pregnancy; q_{ij} is the normalized lead dose for race j; t_{ij} is the gestational day on which race j occurs; and $\delta(\cdot)$ is the Dirac delta function. Each race therefore contributes one exposure pulse on race day.

Let $B(t)$, $S(t)$, and $K(t)$ denote lead in blood, soft tissue, and bone, respectively.[4] The dynamics of lead across the three compartments are given by

$$\dot{B}(t) = E(t) + k_{KB}(t)K(t) + k_{SB}S(t) - (k_{BS} + k_{BK} + k_{B0})B(t) \tag{8}$$
$$\dot{S}(t) = k_{BS}B(t) - k_{SB}S(t) \tag{9}$$
$$\dot{K}(t) = k_{BK}B(t) - k_{KB}(t)K(t) \tag{10}$$

where $E(t)$ is the race-day lead input entering maternal blood on gestational day t, equal to the dose q_{ij} on the day of a race and zero otherwise; k_{B0} is the rate at which blood lead is eliminated through excretion; k_{BS} and k_{SB} are the rates of transfer between blood and soft tissue; and k_{BK} and $k_{KB}(t)$ are the rates of transfer between blood and bone.

[4]For ease of notation, I suppress the subscript i.

Lead stored in maternal bone can be released into the blood during bone resorption. Maternal bone turnover changes over gestation, and skeletal lead can contribute to maternal blood lead, particularly later in pregnancy (Téllez-Rojo et al., 2004; Manton et al., 2003). Because the model simplifies the O'Flaherty framework into a single bone compartment, I approximate the bone-to-blood transfer rate, $k_{KB}(t)$, to increase over gestation. I parameterize $k_{KB}(t)$ as

$$k_{KB}(t) = \begin{cases} k_{KB}^{(1)} = 0.0003 & 0 \leq t \leq 97 \\ k_{KB}^{(2)} = 0.00045 & 98 \leq t \leq 195 \\ k_{KB}^{(3)} = 0.0006 & 196 \leq t \leq 280 \end{cases} \tag{11}$$

The values of $k_{KB}(t)$ are calibrated parameters to capture the increase in skeletal lead mobilization over pregnancy. I set the baseline rate to $k_{KB} = 0.0003$ per day, approximately the daily equivalent of a 10 percent annual skeletal turnover rate (Manolagas and Parfitt, 2010). I allow this rate to increase with each trimester to account for changes in maternal bone resorption during pregnancy. In particular, Curtis et al. (2021) examine C-terminal telopeptide of type I collagen (CTX), a biochemical marker of bone resorption that is released during the breakdown of type I collagen in bone. They find that median maternal urinary CTX approximately doubled between 14 and 34 weeks of gestation. Therefore, I scale the baseline bone-to-blood transfer rate by 1, 1.5, and 2 across the first, second, and third trimesters, respectively, yielding $k_{KB} = 0.00030$, 0.00045, and 0.00060 per day. The remaining transfer rates are also calibrated using the human stable-isotope tracer study of Rabinowitz et al. (1976), which estimates lead movement across three compartments corresponding approximately to blood, soft tissue, and bone. The EPA translates their estimates into daily transfer rates of approximately 0.0079 from blood to soft tissue, 0.0033 from soft tissue back to blood, 0.0037 from blood to bone, and 0.019 from blood to excretion (U.S. Environmental Protection Agency, 2001). Hence, I set the parameters as $k_{BS} = 0.0079$, $k_{SB} = 0.0033$, $k_{BK} = 0.0037$, and $k_{B0} = 0.019$ per day. Solving the system using the Euler method with a one-day time step, I set the initial conditions $B(0) = S(0) = K(0) = 0$ and simulate lead levels from gestational day 0 to day 280.[5] The system produces a daily maternal blood-lead trajectory, $B_i(t)$, which I use as a proxy for fetal lead exposure given that lead readily crosses the placenta and lead levels in maternal and umbilical cord blood are highly correlated (Zhu et al., 2010).

To construct trimester-specific exposure measures, I average $B_i(t)$ over the gestational

[5]The state variables capture lead attributable to lead exposure on race days rather than the mother's total lead burden.

days in each trimester. Note that $B_i(t)$ is expressed in normalized exposure units rather than micrograms per deciliter. Let T_k denote the set of gestational days in trimester $k \in \{1, 2, 3\}$, where $T_1 = \{0, \ldots, 97\}$, $T_2 = \{98, \ldots, 195\}$, and $T_3 = \{196, \ldots, 280\}$, and let $|T_k|$ denote the number of days in T_k. Mean predicted maternal blood lead for mother i in trimester k is

$$\overline{B}_{ik} = \frac{1}{|T_k|} \sum_{t \in T_k} B_i(t) \tag{12}$$

To measure cumulative exposure over pregnancy, I sum $B_i(t)$ across gestational days to obtain the area under the maternal blood-lead trajectory (AUC):

$$\text{AUC}_i = \sum_{t=1}^{280} B_i(t). \tag{13}$$

I also measure peak exposure as the maximum daily maternal blood-lead level over pregnancy:

$$\text{Peak}_i = \max_{t \in \{1,\ldots,280\}} B_i(t) \tag{14}$$

Hence, the trimester means $\overline{B}_{i1}, \overline{B}_{i2}, \overline{B}_{i3}$, the cumulative-dose measure AUC_i, and the peak Peak_i describe the simulated maternal blood lead trajectory based on when the exposure occurred, how much lead was accumulated overall, and how high it spiked.

Figure 4 plots the simulated blood-lead trajectories for mothers within 25 km of TMS before the transition, grouped by the trimesters in which they were exposed to a leaded race. The top panel shows mothers exposed in all three trimesters, whose average blood lead never returns to zero because lead from earlier races has not cleared before the next one. The rest of the panels show exposure in a single trimester. For mothers exposed only in the first trimester, predicted blood lead rises near the end of the first trimester and declines gradually through the second, and it has not fully cleared by birth. Mothers exposed only in the second trimester show the same pattern one trimester later, with lead carrying into the third trimester. Exposure only in the third trimester is rare because TMS held races in the spring, early summer, and fall, so pregnancies almost always include a race before the third trimester. Although about 19,800 mothers were exposed to a leaded race in the third trimester, nearly all were also exposed earlier in pregnancy. Being exposed only in the third trimester requires 196 consecutive days without a race before the third trimester begins, and the only such gap at TMS is the 197 days between September

13, 2002 and March 29, 2003. The nine mothers in this group were therefore all conceived on September 14, 2002. In each case, the PBPK trimester means attribute part of a race's lead to the trimesters that follow it.

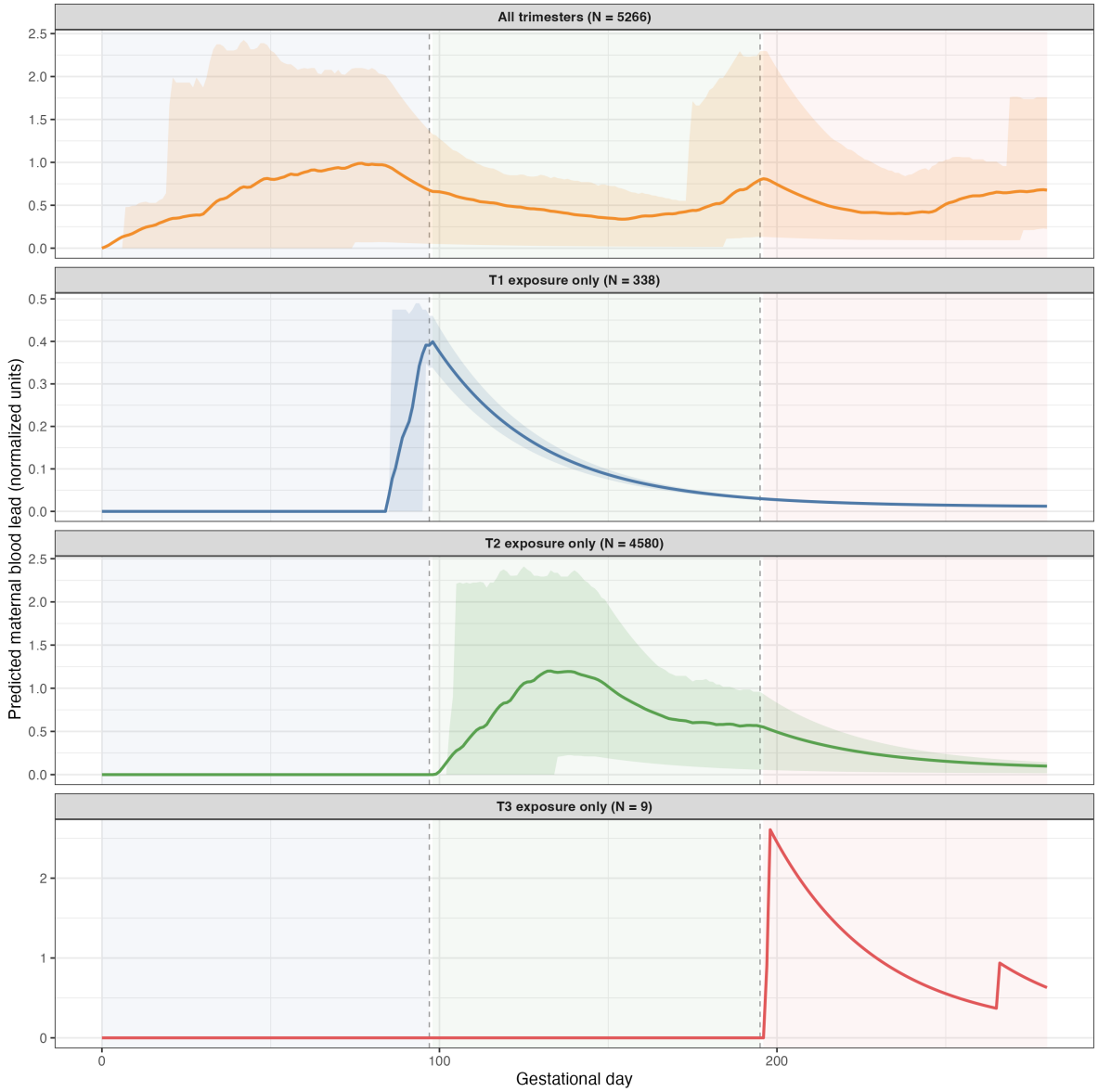


Figure 4. The panels show mothers living within 25 km of TMS whose only leaded race exposures occurred in all trimesters and each trimester. The first panel shows mothers exposed in all trimesters ($N = 5{,}266$) while the second panel shows mothers exposed only in the first trimester ($N = 338$). The third and fourth panels show mothers exposed only in the second trimester ($N = 4{,}580$) and the third trimester ($N = 9$), respectively.

5 Empirical Strategy

5.1 Within-TMS Identification

The identification strategy relies on comparing births conceived before and after the 2006/2007 switch to unleaded gasoline at TMS while exploiting variation in the timing, location, and intensity of race-day exposure. The level of emissions released during a race varies with factors such as accidents and race length and is plausibly unrelated to maternal health. Appendix A tests whether log emissions in each trimester are correlated with predetermined maternal characteristics: maternal age, marital status, WIC enrollment, race, first birth, and insurance status. Only the coefficient for WIC enrollment and log emissions in trimester 2 is statistically significant and at the 10% level whereas the rest are statistically insignificant.

Since all mothers live within 25 km of TMS and experience at least one race during pregnancy, there is no control group in the traditional sense. Exposure varies across mothers with how many races and how much lead fall within their pregnancies and within each trimester. Exploiting variation in conception timing relative to the fixed calendar of TMS races, I estimate the effect of exposure over the full pregnancy, which is the main specification, and of exposure in each trimester, separately for the pre- and post-transition periods. If the pre-transition relationship is driven by lead exposure, the estimated effects should be negative before the transition and attenuate toward zero after the transition. This pattern would support the interpretation that the pre-transition effects are driven by lead rather than by other factors associated with racing.

Additionally, the birth-month and birth-year fixed effects remove differences across years and across calendar months but not the race calendar of a given year. Therefore, the change in the race schedule from one year to the next and the timing of conception among mothers delivering in the same month and year form two sources of variation. Mothers who deliver infants in the same month might have conceived on different dates so their pregnancies also start and end on different dates. Therefore, a race that happens near the start or end of pregnancy might fall inside some mothers' 280-day pregnancy windows and outside others'. For example, consider two mothers who both deliver at 40 weeks in April 2005, one on April 5 and the other on April 25. The first mother's pregnancy runs from early July 2004 to April 5, 2005, and includes one race in October 2004. The second mother's pregnancy starts and ends about three weeks later so it includes the same October race and also the two races of the April 16-17 race weekend. Because the two mothers deliver in the same month of the same year, the fixed effects treat them alike, but their exposure differs because the April race weekend falls after the first mother delivers and before the second mother delivers. Appendix D measures how much each source contributes. Before the transition, about one third of the variation in race counts

that remains after the fixed effects comes from the change in the schedule, and two thirds comes from the timing of conception. That timing variation is split about equally between races near the start of the pregnancy and races near its end. After the transition, the schedule did not change so almost all of the variation comes from timing. Once the fixed effects are included, the number of races explains 99.7 percent of the variation in log emissions before the transition and 96.8 percent after so log emissions and race counts capture essentially the same variation. The pre-transition estimates, however, disappear when conception year-by-month fixed effects replace the birth-year and birth-month fixed effects (Appendix F).

I use birth-year and birth-month fixed effects as the baseline because they keep the variation from when each pregnancy started relative to the race calendar. The log-emission estimates remain large with birth year-by-month fixed effects so shocks common to births in a given month do not explain them (Appendix D). Conception year-by-month fixed effects instead compare only mothers who conceived in the same month. Because the month of conception largely determines which races fall in a pregnancy, they remove about two-thirds of the exposure variation and leave only mothers who conceived a few days apart (Table 17). Gestational length explains less than 1 percent of the baseline variation, although this does not rule out a mechanical relationship between exposure and gestational length. The balance test shows no relationship between exposure and observed maternal characteristics (Table 8), but it cannot rule out unobserved differences. Furthermore, the estimates with conception fixed effects become positive and statistically significant when the exposure window ends at day 195 before the transition (Table 16). This is a limitation of the design as the data cannot establish whether these fixed effects remove meaningful variation or confounding.

5.2 Baseline Specification

The baseline reduced-form specification is estimated separately for the pre- and post-transition samples. The primary outcome is birthweight in grams, and secondary outcomes include gestational age in weeks, low birthweight status ($<$ 2,500 grams), and birthweight conditional on gestational age. All outcome variables are estimated on log emissions ($\widetilde{E}_i$ and $\widetilde{E}_{ik}$) and race counts (R_i and R_{ik}) as the first two primary exposure measures. For the rest of the exposure measures such as distance, wind direction, and PBPK model, I estimate only for the primary outcome variable which is birthweight. The following reduced-form specification estimates each outcome on log emissions over full pregnancy.

$$Y_{ict} = \alpha + \beta\, Z_i + X'_{ict}\gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \tag{15}$$

where Y_{ict} is the outcome variable for birth i in census tract c at calendar time t; Z_i is the log emissions over full pregnancy ($\widetilde{E}_i$); X_{ict} is a vector of maternal demographic, socioeconomic, and health controls (marital status, WIC enrollment, education, race, ethnicity, first birth, insurance status, and maternal age group, chronic hypertension, previous preterm birth, smoking during pregnancy, pre-pregnancy diabetes, previous cesarean delivery, previous poor pregnancy outcome, and infertility treatment)[6]; μ_c are census tract fixed effects; μ_y are birth-year fixed effects; and μ_m are birth-month fixed effects. Standard errors are clustered at the census tract level.

I also estimate a joint trimester specification,

$$Y_{ict} = \alpha + \sum_{k=1}^{3} \beta_k\, Z_{ik} + X'_{ict}\gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \tag{16}$$

where Z_{ik} is the log emissions in trimester k ($\widetilde{E}_{ik}$ in Equation 3). Because trimester log emissions are zero for about a third of mothers in each trimester and take similar values among mothers with at least one race in that trimester, the trimester log-emission coefficients largely compare mothers with and without a race in that trimester. Appendix C shows that a single linear term fits this pattern poorly, so I rely on race counts for the trimester results.

The race-count estimates use the same specifications,

$$Y_{ict} = \alpha + \beta\, R_i + X'_{ict}\gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \tag{17}$$

where R_i is the number of races during mother i's pregnancy. Similarly, the trimester version for race counts replaces Z_{ik} in Equation 16 with R_{ik}, which is the number of races during trimester k. I estimate Equation (15) and Equation (17) separately for the pre- and post-transition samples. Before the transition, it measures the lead actually emitted

[6]WIC enrollment and insurance status are not recorded on the birth certificate before 2005, and neither are pre-pregnancy diabetes, previous cesarean delivery, previous poor pregnancy outcome, and infertility treatment. For births in 2003-04, I set these variables to zero and include indicators for missing values, which are absorbed by the birth-year fixed effects. The coefficients on these controls are therefore identified from births in 2005-06, and births in 2003-04 remain in the sample.

during races. After the transition, it measures the amount of lead that would have been emitted if the race had used the same leaded fuel as before the 2006/2007 switch. The pre-transition β captures the relationship between lead emissions during pregnancy and birth outcomes, whereas the post-transition β captures the relationship between the corresponding counterfactual emissions measure and birth outcomes after races no longer emitted lead.

To test whether the distance gradient differs across periods, I pool the pre- and post-transition samples and interact distance from TMS with a pre-transition indicator:

$$\text{BW}_{ict} = \alpha + \delta\,\text{Dist}_c + \lambda\,\text{Pre}_i + \phi\,(\text{Dist}_c \times \text{Pre}_i) + X'_{ict}\gamma + \mu_y + \mu_m + \varepsilon_{ict} \tag{18}$$

where Dist_c is the distance in kilometers from the centroid of mother i's census tract c to TMS; Pre_i is an indicator equal to one if the birth was conceived before the transition; X_{ict} is the same vector of controls as in Equation (15); and μ_y and μ_m are birth-year and birth-month fixed effects. Census tract fixed effects are omitted because they absorb distance. Standard errors are clustered at the census tract level. The coefficient δ is the distance gradient after the transition, $\delta + \phi$ is the gradient before the transition, and ϕ measures the difference between the two.

Similarly, the specification for wind direction is:

$$\text{BW}_{ict} = \alpha + \theta\,\text{ShareDownwind}_i + \rho\,R_i + X'_{ict}\gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \tag{19}$$

where BW_{ict} is birthweight in grams for birth i in census tract c at calendar time t; ShareDownwind_i is the fraction of race days during mother i's pregnancy on which she was downwind of TMS (Equation (4)); R_i is the number of races at TMS during her pregnancy; X_{ict} is the same vector of controls as in Equation (15); μ_c are census tract fixed effects; μ_y are birth-year fixed effects; and μ_m are birth-month fixed effects. Holding the number of races during pregnancy fixed, θ measures the difference in birthweight associated with an increase in the share of race days on which the mother was downwind of TMS. A negative θ indicates that mothers who were downwind on a larger share of race days experienced greater plume exposure and had infants with lower birthweight. I also estimate Equation (19) replacing ShareDownwind_i with AnyDownwind_i (Equation (5)).

Finally, I estimate the relationship between simulated maternal blood lead from the PBPK

model and birthweight:

$$\text{BW}_{ict} = \alpha + \sum_{k=1}^{3} \pi_k \overline{B}_{ik} + X'_{ict}\gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \tag{20}$$

where $\overline{B}_{ik}$ is mean predicted maternal blood lead for mother i in trimester k; π_k is the change in birthweight associated with a one-unit increase in mean blood lead in trimester k; and the remaining terms are defined as in Equation (15). For the full pregnancy, I replace the three trimester means with either cumulative blood lead (AUC_i) or peak blood lead (Peak_i). Because blood lead from racing is zero after the transition, I estimate Equation (20) for the pre-transition sample only.

6 Results

Table 1 presents summary statistics for live births within 25 km of TMS for the pre- and post-transition periods. Average birthweight is 3,358 grams in the pre-transition period and 3,339 grams in the post-transition period while the proportion of low birthweight infants increases slightly from 4.7% to 5.1%. Average gestational age remains relatively stable at approximately 38.5 weeks in both periods. Maternal characteristics are also different across the two periods. The proportion of married mothers decreases from 80.6% to 75.2% while WIC enrollment increases from 24.6% to 26.8%. The average distance between maternal residence and TMS is approximately 19 km in both periods. The average number of races occurring during each trimester is higher in the post-transition period.

Table 1. Summary Statistics: Births Within 25 km of TMS

	Pre-transition		Post-transition		Full sample	
	mean	sd	mean	sd	mean	sd
Birthweight (g)	3358.3551	(522.4517)	3339.0232	(534.8890)	3348.0461	(529.6457)
Low birthweight (<2500g)	0.0465	(0.2105)	0.0513	(0.2206)	0.0493	(0.2165)
Gestational age (weeks)	38.4891	(1.7654)	38.4709	(1.7923)	38.4799	(1.7791)
Apgar score (5 min)	8.9090	(0.5417)	8.8794	(0.6524)	8.8940	(0.6066)
Gestational hypertension	0.0428	(0.2023)	0.0431	(0.2031)	0.0435	(0.2040)
Eclampsia	0.0013	(0.0361)	0.0006	(0.0236)	0.0009	(0.0292)
Gestational diabetes	0.0220	(0.1467)	0.0357	(0.1856)	0.0299	(0.1702)
Maternal age (years)	28.7219	(5.8379)	28.5265	(5.8846)	28.6021	(5.8581)
Married	0.8058	(0.3956)	0.7522	(0.4318)	0.7786	(0.4152)
Education	0.4079	(0.4914)	0.3854	(0.4867)	0.3942	(0.4887)
WIC enrollment	0.2458	(0.4306)	0.2675	(0.4427)	0.2584	(0.4377)
Black	0.0457	(0.2088)	0.0537	(0.2254)	0.0499	(0.2178)
White	0.8757	(0.3300)	0.8539	(0.3532)	0.8609	(0.3461)
Hispanic	0.1986	(0.3989)	0.2330	(0.4228)	0.2155	(0.4112)
First birth	0.4053	(0.4910)	0.4096	(0.4918)	0.4084	(0.4915)
Insured	0.8874	(0.3161)	0.8904	(0.3125)	0.8864	(0.3174)
Chronic hypertension	0.0091	(0.0949)	0.0069	(0.0829)	0.0082	(0.0899)
Previous preterm birth	0.0068	(0.0823)	0.0092	(0.0952)	0.0080	(0.0893)
Smoked during pregnancy	0.0479	(0.2135)	0.0392	(0.1941)	0.0447	(0.2067)
Pre-pregnancy diabetes	0.0050	(0.0705)	0.0034	(0.0584)	0.0040	(0.0631)
Previous cesarean	0.1346	(0.3413)	0.1425	(0.3496)	0.1396	(0.3466)
Previous poor pregnancy outcome	0.0058	(0.0758)	0.0023	(0.0483)	0.0033	(0.0572)
Infertility treatment	0.0095	(0.0971)	0.0055	(0.0739)	0.0067	(0.0815)
Distance to TMS (km)	19.1642	(4.5994)	18.7322	(4.9143)	18.9299	(4.7695)
Races during T1	1.1538	(1.0333)	1.6225	(1.2654)	1.4081	(1.1793)
Races during T2	1.2221	(1.0656)	1.5863	(1.2688)	1.4264	(1.1864)
Races during T3	1.0726	(1.0290)	1.4133	(1.2669)	1.2566	(1.1873)
Mean maternal blood lead, T1	0.2681	(0.3305)	0.0000	(0.0000)	0.1610	(0.2847)
Mean maternal blood lead, T2	0.4304	(0.3258)	0.0000	(0.0000)	0.2371	(0.3164)
Mean maternal blood lead, T3	0.4952	(0.3634)	0.0000	(0.0000)	0.2470	(0.3434)
AUC, full pregnancy	110.5406	(55.7561)	0.0000	(0.0000)	60.0046	(66.8078)
Peak blood lead	2.4448	(0.9472)	0.0000	(0.0000)	1.3237	(1.3541)
Observations	31362		28620		71366	

Notes: Mean with standard deviation in parentheses. Pre-transition is defined as conception before December 17, 2005; post-transition as conception on or after March 1, 2007. Full sample includes all births in the study area (including the transition window excluded from the pre/post columns). Sample consists of all Texas births from 2003–2010 whose nearest racetrack is TMS and who reside within 25 km. Education is defined as having at least a college degree.

6.1 Pre-Transition vs. Post-Transition Effect

Table 2 reports the estimated effects of prenatal exposure to race activity at TMS on birthweight (grams), low birthweight (<2,500 grams), gestational age (weeks), and birthweight conditional on gestational age (grams). Each specification is estimated separately for the pre- and post-transition periods. Because racing continued after the transition without leaded fuel, the post-transition period serves as a placebo test.

Panel A reports the estimates using the logarithm of lead emissions. Before the transition, a doubling of lead emitted during pregnancy reduces birthweight by 26.8 grams ($-38.72 \times \ln(2)$), shortens gestation by 0.14 weeks or about one day ($-0.206 \times \ln(2)$), and raises the probability of low birthweight by 0.73 percentage points ($0.0106 \times \ln(2)$), which is a 16 percent increase relative to the pre-transition prevalence of 4.7 percent. All three estimates are statistically significant at the one percent level. After the transition, the corresponding estimates are small and statistically insignificant. Birthweight conditional on gestational age is not statistically significant in either period (-2.03 grams before and -15.51 grams after transition), which is consistent with a pathway through shorter gestation rather than slower fetal growth. The trimester log-emission estimates are small and mostly statistically insignificant in both periods. As discussed in Section 5, they largely compare mothers with and without a race in a given trimester so I use race counts for the trimester results (Appendix C).

Panel B reports the estimates based on the number of races during pregnancy. Before the transition, each additional race reduces birthweight by 14.5 grams, shortens gestation by 0.07 weeks or about half a day, both significant at the one percent level. In addition, each additional race lowers birthweight in all trimesters, with the largest reduction in the second trimester. One additional race in the second trimester reduces birthweight by 32.6 grams or about one percent of the average birthweight before the transition. After the transition, the full pregnancy and trimester estimates are substantially smaller and statistically insignificant.

Similarly, each additional race in the second trimester raises the probability of low birthweight by 1.15 percentage points, which is a 25 percent increase relative to the pre-transition prevalence of 4.7 percent. However, the estimate remains positive and statistically significant after the transition which is not consistent with the lead driving the effect. For the full pregnancy, the race count does not affect low birthweight in either period. Across all three trimesters, each additional race decreases gestational age, especially in the second trimester by approximately one day, while the estimates are close to zero after the transition. After the transition, each additional race in the first trimester is associated with a reduction in birthweight conditional on gestational age by 9.4 grams while there is no effect before the transition. Appendix G shows that the shorter gestation reflects a higher risk of preterm birth. A doubling of the lead emitted during pregnancy raises the probability of preterm birth (less than 37 weeks) by 1.6 percentage points, a 21 percent increase relative to the pre-transition prevalence of 7.8 percent, and the probability of early preterm birth (less than 34 weeks) by 0.7 percentage points with no effect after the transition. The next subsection examines whether these relationships differ across

cohorts with more race exposure.

Table 2. Main Reduced-Form Results

Panel A: Log emissions								
	Pre-transition (leaded)				Post-transition (as-if-leaded, placebo)			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Full pregnancy	BW (g)	LBW	Gest. age (wks)	BW | GA	BW (g)	LBW	Gest. age (wks)	BW | GA
Log emissions, full pregnancy	-38.7199***	0.0106***	-0.2064***	-2.0346	0.1794	0.0003	0.0877	-15.5137
	(9.6535)	(0.0039)	(0.0471)	(4.8255)	(19.6126)	(0.0095)	(0.0823)	(11.7159)
Observations	31353	31353	31362	31353	28620	28620	28620	28620
R-squared	0.0374	0.0163	0.0246	0.3516	0.0405	0.0138	0.0242	0.3911
By trimester (T1+T2+T3 joint)								
Log emissions, T1	2.1736*	-0.0010*	0.0160***	-0.3288	-2.0111	0.0007	0.0031	-2.5675***
	(1.2354)	(0.0005)	(0.0046)	(0.8577)	(1.2790)	(0.0007)	(0.0056)	(0.7793)
Log emissions, T2	0.1193	-0.0003	0.0016	-0.0710	-0.6628	0.0009	0.0029	-1.1814
	(1.3419)	(0.0006)	(0.0059)	(1.0081)	(1.4632)	(0.0006)	(0.0059)	(0.9622)
Log emissions, T3	-1.4274	-0.0005	-0.0090	0.1497	0.5444	-0.0001	0.0078*	-0.8565
	(1.5505)	(0.0006)	(0.0060)	(1.0616)	(1.3006)	(0.0006)	(0.0043)	(0.8951)
Observations	31353	31353	31362	31353	28620	28620	28620	28620
R-squared	0.0368	0.0162	0.0236	0.3516	0.0406	0.0139	0.0243	0.3913
Panel B: Cumulative race count								
	BW (g)	LBW	Gest. age (wks)	BW | GA	BW (g)	LBW	Gest. age (wks)	BW | GA
Full pregnancy								
Total Races, Full Pregnancy	-14.4901***	0.0024	-0.0725***	-1.3528	-1.8793	0.0013	0.0154	-4.6310
	(4.6666)	(0.0018)	(0.0206)	(2.8761)	(4.5715)	(0.0024)	(0.0204)	(2.9122)
Observations	31353	31353	31362	31353	28620	28620	28620	28620
R-squared	0.0371	0.0160	0.0237	0.3516	0.0405	0.0138	0.0242	0.3911
By trimester (T1+T2+T3 joint)								
Races during T1	-16.7808**	0.0040*	-0.0658**	-3.8169	-7.7685	0.0036	0.0090	-9.3738***
	(6.3906)	(0.0023)	(0.0286)	(3.5617)	(6.4732)	(0.0032)	(0.0280)	(3.4240)
Races during T2	-32.6293***	0.0115***	-0.1693***	-2.6193	-6.2113	0.0074**	-0.0173	-3.1173
	(6.6665)	(0.0027)	(0.0364)	(4.1654)	(7.0154)	(0.0030)	(0.0290)	(4.7801)
Races during T3	-18.1562***	0.0038*	-0.1069***	0.2097	1.6833	0.0018	0.0067	0.4871
	(5.4905)	(0.0021)	(0.0241)	(4.0145)	(5.4976)	(0.0028)	(0.0195)	(4.3901)
Observations	31353	31353	31362	31353	28620	28620	28620	28620
R-squared	0.0375	0.0167	0.0250	0.3516	0.0406	0.0141	0.0243	0.3912

Standard errors in parentheses. Sample: TMS live births within 25 km. All regressions include census tract and year and month fixed effects plus standard maternal controls. Standard errors clustered by tract. Outcomes: birthweight in grams (BW), low birthweight <2500g (LBW), gestational age in weeks, and birthweight conditional on gestational age (BW | GA). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 tests whether the estimates before and after the transition are statistically different. I estimate a single regression on the pre- and post-transition samples and fully interacts every regressor, control, and fixed effect with a pre-transition indicator. Each coefficient in the table is the difference between the pre- and post-transition estimates in Table 2. Standard errors are clustered by census tract. Over the full pregnancy, a one-unit increase in log emissions lowers birthweight by 38.9 grams more before the transition than after, but this difference is significant only at the 10 percent level, mainly because the post-transition estimate is imprecise (standard error 23.1). It also shortens gestation by 0.29

weeks more before the transition, a difference significant at the 1 percent level. With race counts, each race lowers birthweight by 12.6 grams more before the transition (significant at the 10 percent level), and by 26.4 and 19.8 grams more for races in the second and third trimesters (both significant at the 5 percent level). The differences for low birthweight and for birthweight conditional on gestational age are not statistically significant. The test therefore gives suggestive evidence for birthweight, the main result, and clearer evidence for gestational age.

Table 3. Difference Between Pre- and Post-Transition Estimates

	(1) BW (g)	(2) LBW	(3) Gest. age (wks)	(4) BW | GA
Panel A: Log emissions *Full pregnancy*				
Log emissions, full pregnancy	-38.8993* (23.1436)	0.0103 (0.0096)	-0.2941*** (0.0923)	13.4791 (13.0128)
Observations *By trimester (T1+T2+T3 joint)*	59973	59973	59982	59973
Log emissions, T1	4.1847** (2.0809)	-0.0017* (0.0009)	0.0129 (0.0082)	2.2387* (1.2222)
Log emissions, T2	0.7821 (1.6928)	-0.0012 (0.0009)	-0.0013 (0.0080)	1.1104 (1.3796)
Log emissions, T3	-1.9718 (2.1087)	-0.0004 (0.0008)	-0.0169** (0.0076)	1.0062 (1.5201)
Joint test, p-value Observations	0.125 59973	0.331 59973	0.037 59982	0.276 59973
Panel B: Cumulative race count *Full pregnancy*				
Total races, full pregnancy	-12.6109* (6.9615)	0.0011 (0.0028)	-0.0879*** (0.0288)	3.2782 (4.1488)
Observations *By trimester (T1+T2+T3 joint)*	59973	59973	59982	59973
Races during T1	-9.0123 (10.0869)	0.0003 (0.0037)	-0.0748* (0.0421)	5.5569 (5.4392)
Races during T2	-26.4180** (9.9821)	0.0040 (0.0037)	-0.1520*** (0.0472)	0.4980 (6.8366)
Races during T3	-19.8395** (7.9480)	0.0020 (0.0031)	-0.1136*** (0.0276)	-0.2774 (6.1539)
Joint test, p-value Observations	0.046 59973	0.690 59973	0.001 59982	0.755 59973

Each reported coefficient is the exposure measure interacted with a pre-transition indicator, which equals the pre-transition estimate minus the post-transition estimate in Table 2, from a regression that pools both periods and interacts the exposure measures, controls, and census tract, birth-year and birth-month fixed effects with a pre-transition indicator. Standard errors clustered by tract across the pooled sample in parentheses. The joint test is that all three trimester differences are zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Heterogeneity by Birth Cohort

Pre 2005, TMS hosted one major race weekend each year, typically in the spring, along with two smaller truck races. Beginning in 2005, TMS added a second Cup and Xfinity

race weekend in the fall so I examine whether the results differ between births before or after 2005.

Table 4 re-estimates the pre-transition specifications separately for births in 2003-04 and for births from 2005-06. Among births from 2005-06, the estimated effects are larger in magnitude than those in the pooled pre-transition sample. A doubling of the lead emitted during pregnancy reduces birthweight by 55.8 grams (1.7 percent of the mean), raises the probability of low birthweight by 1.72 percentage points (a 36 percent increase relative to the 4.8 percent prevalence in these cohorts), and shortens gestation by 0.27 weeks (about 1.9 days). Each additional race during pregnancy reduces birthweight by 24.3 grams, and one additional race during the second trimester reduces birthweight by 52.8 grams, raises the probability of low birthweight by 1.74 percentage points, and shortens gestation by 0.27 weeks.

In contrast, the estimates for births in 2003 and 2004 are close to zero. This is not because there is too little data to detect an effect in these years. Exposure varies about as much as it does for later births. Appendix E shows that the estimates for the 2005 and later cohorts remain large when birth year-by-month fixed effects are added so the variation from the schedule change does not explain the difference between the two cohorts (Table 15). The difference still remains large when the regressions exclude pregnancies that include the new fall race weekend, keep only the controls recorded in both cohorts, restrict the sample to births whose gestation agrees, or date conception using the obstetric estimates. Appendix C shows that this difference is not a result of the linear functional form. I cannot explain this difference, and the main results should be read as driven by the 2005 and later cohorts.

Table 4. Pre-Transition Results by Birth Year

Panel A: Log emissions								
	Born in 2003–04				Born in 2005–06			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Full pregnancy	BW (g)	LBW	Gest. age (wks)	BW | GA	BW (g)	LBW	Gest. age (wks)	BW | GA
Log emissions, full pregnancy	6.9650	-0.0041	-0.0200	13.2999	-80.5114***	0.0248***	-0.3830***	-14.5553*
	(17.3471)	(0.0075)	(0.0747)	(10.8532)	(10.8541)	(0.0058)	(0.0575)	(7.4464)
Observations	16135	16135	16143	16135	15218	15218	15219	15218
R-squared	0.0411	0.0181	0.0188	0.3457	0.0410	0.0200	0.0407	0.3622
By trimester (T1+T2+T3 joint)								
Log emissions, T1	2.1285	-0.0002	0.0144	0.3255	2.3884*	-0.0017**	0.0205***	-1.1557
	(2.2218)	(0.0010)	(0.0097)	(1.5177)	(1.3607)	(0.0008)	(0.0061)	(0.9643)
Log emissions, T2	3.4833*	-0.0003	0.0135	1.3620	-3.3192	0.0001	-0.0120	-1.2452
	(2.0443)	(0.0008)	(0.0091)	(1.4314)	(2.0219)	(0.0011)	(0.0080)	(1.5496)
Log emissions, T3	1.7215	-0.0012	0.0014	1.5603	-4.6763**	0.0001	-0.0193**	-1.3261
	(2.3608)	(0.0010)	(0.0102)	(1.2577)	(1.8636)	(0.0009)	(0.0074)	(1.4792)
Observations	16135	16135	16143	16135	15218	15218	15219	15218
R-squared	0.0414	0.0183	0.0193	0.3457	0.0390	0.0190	0.0372	0.3622
Panel B: Cumulative race count								
Full pregnancy	BW (g)	LBW	Gest. age (wks)	BW | GA	BW (g)	LBW	Gest. age (wks)	BW | GA
Total Races, Full Pregnancy	6.4072	-0.0038	0.0129	6.7699	-24.2873***	0.0043	-0.1179***	-3.9774
	(11.0326)	(0.0049)	(0.0440)	(6.5405)	(5.3848)	(0.0029)	(0.0311)	(3.7637)
Observations	16135	16135	16143	16135	15218	15218	15219	15218
R-squared	0.0411	0.0181	0.0188	0.3456	0.0394	0.0186	0.0375	0.3621
By trimester (T1+T2+T3 joint)								
Races during T1	6.6983	-0.0017	0.0511	4.3238	-26.5365***	0.0059*	-0.1127***	-7.1659*
	(13.4974)	(0.0057)	(0.0681)	(8.4652)	(6.7503)	(0.0034)	(0.0332)	(3.7108)
Races during T2	10.5177	-0.0004	-0.0026	14.0067	-52.7733***	0.0174***	-0.2677***	-6.6772
	(16.1685)	(0.0069)	(0.0813)	(9.9987)	(8.0472)	(0.0040)	(0.0384)	(6.2571)
Races during T3	7.8495	-0.0044	-0.0291	12.0668	-30.6827***	0.0067**	-0.1679***	-1.7267
	(14.7845)	(0.0063)	(0.0664)	(8.0394)	(6.9452)	(0.0032)	(0.0340)	(5.8158)
Observations	16135	16135	16143	16135	15218	15218	15219	15218
R-squared	0.0412	0.0182	0.0191	0.3457	0.0407	0.0202	0.0411	0.3622

Standard errors in parentheses. TMS held one Cup race weekend a year (in spring) until a second, fall Cup weekend was added in November 2005. All regressions include census tract, birth-year, and birth-month fixed effects plus standard maternal controls. Standard errors clustered by tract. Outcomes: birthweight in grams (BW), low birthweight <2500g (LBW), gestational age in weeks, and birthweight conditional on gestational age (BW | GA). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3 Spatial Gradient Test

If airborne lead is driving the relationship, birthweight should increase with distance from the track in the pre-transition period as lead concentrations decay with distance, while the birthweight gradient should be approximately flat after the transition. Figure 5 shows that predicted birthweight increases with distance from TMS before the transition. This is consistent with the fact that mothers who reside closer to TMS are exposed to higher concentrations of lead and deliver infants with lower birthweight. Once lead is removed from the fuel, the distance gradient becomes much flatter and slightly downward sloping. This pattern is consistent with lead exposure driving the pre-transition

relationship rather than other race-related factors that persisted after the transition.

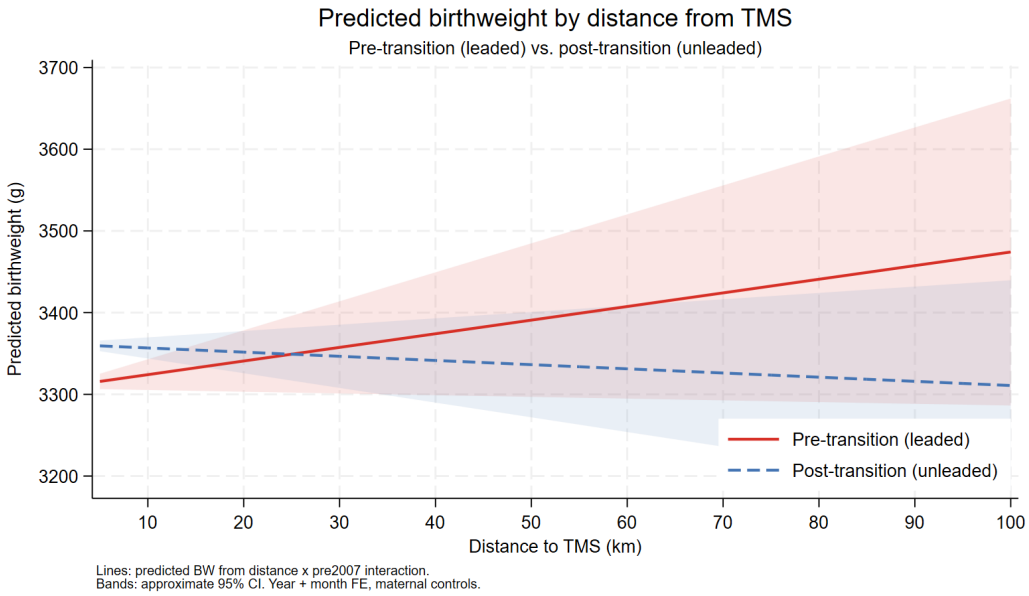


Figure 5. Predicted birthweight by distance from TMS. (Pre-transition = solid red; Post-transition = dashed blue). Birthweight is regressed on distance $\times$ pre-transition with birth-year and birth-month fixed effects and maternal controls. Shaded regions are approximate 95% confidence intervals.

Before the transition, birthweight increases by 1.67 grams for each additional kilometer from the track (significant at the 1 percent level). After the transition, the distance gradient is -0.51 grams per kilometer and statistically insignificant. The difference between the two slopes, given by the interaction between the pre-transition indicator and distance, is 2.18 grams per kilometer and is significant at the 5 percent level. Across the interquartile range of distance in the pooled sample (15.5 to 22.5 kilometers), the pre-transition gradient implies that infants born to mothers living farther from the track weighed about 12 grams more than those born to mothers living closer to it.

6.4 Wind Direction

Wind direction is plausibly exogenous to maternal sorting around the racetrack. Hence, mothers who reside downwind of the track on race days should experience larger birthweight reductions than those upwind, holding fixed the number of races during pregnancy. Table 5 reports the relationship between downwind exposure over the full pregnancy and birthweight. Panel A uses the share of race days during pregnancy on which the mother was downwind while Panel B uses a binary indicator for whether the mother was downwind on at least one race day during pregnancy. All specifications control for

the number of races during pregnancy so that the wind coefficients compare mothers with the same number of races who differ only in wind direction on those days.

Before the transition, mothers who were downwind on every race day during pregnancy delivered infants weighing 39.8 grams less, on average, than mothers who were never downwind (significant at the 5 percent level). Similarly, Panel B shows that being downwind on at least one race day is associated with a 25.3-gram reduction in birthweight, or 0.8 percent of mean birthweight, relative to mothers exposed to the same number of races who were never downwind (significant at the 1 percent level).

After the transition, both downwind estimates are positive and statistically significant (32.7 grams in Panel A and 14.3 grams in Panel B). Appendix H shows that this reflects seasonality rather than race-day exposure. Wind direction on race days differs between the spring and fall race weekends so downwind status partly captures the season of the pregnancy in a given location. When I control for downwind status on non-race days, the post-transition estimate falls to -11.8 grams and is statistically insignificant while the pre-transition estimate grows to -52.7 grams.

Table 5. Wind Direction: Downwind Exposure over the Full Pregnancy and Birthweight

Panel A: Share of race days downwind		
	Pre-transition	Post-transition
	(1)	(2)
Share downwind, full pregnancy	-39.8192**	32.7107*
	(15.1708)	(18.9357)
Total Races, Full Pregnancy	-14.1282***	-1.9385
	(4.5773)	(4.5686)
Observations	31353	28620
R-squared	0.0373	0.0406
Panel B: Any downwind indicator		
	Pre-transition	Post-transition
Any downwind, full pregnancy	-25.2846***	14.3363**
	(8.8857)	(7.0179)
Total Races, Full Pregnancy	-12.7946***	-3.0354
	(4.4447)	(4.6644)
Observations	31353	28620
R-squared	0.0373	0.0406

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km. All regressions include census tract and year and month fixed effects, standard maternal controls, and the number of races during pregnancy (shown). Standard errors clustered by tract. The share downwind measure is the fraction of race days during pregnancy on which the mother's residence was downwind ($\pm 45°$). The Any Downwind indicator equals one if the mother was downwind on at least one race day during pregnancy. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5 PBPK Dose-Response

The reduced-form estimates in Sections 6.1 to 6.4 treat each race as an instantaneous shock and do not allow lead to accumulate in or clear from the mother's body. The PBPK model captures the accumulation by simulating, for each mother, the daily path of blood lead over her full pregnancy. The input dose for each race is normalized by the average lead emitted per leaded race. Simulated blood lead then is based on these doses and on how lead accumulates and is eliminated over pregnancy. The model does not estimate blood lead in absolute units, and I interpret the coefficients by comparing mothers at different points of the simulated distribution. I report the difference in birthweight between mothers at the 25th and 75th percentiles of the exposure distribution; that is, between a mother with relatively low and a mother with relatively high exposure within the middle half of the distribution. Since blood lead from racing is mechanically zero after the transition, the model reports estimates from the pre-transition period only.

Table 6 reports the results. Over the full pregnancy, each additional unit of cumulative blood lead lowers birthweight by 0.65 grams (Panel A) so a mother at the 75th percentile of cumulative exposure delivers an infant about 38 grams lighter than a mother at the 25th percentile (74 to 132 units). One additional unit in peak blood lead also decreases birthweight by 26.6 grams (Panel A). However, peak blood lead is similar across mothers (2.58 to 2.81). Mothers at the 75th percentile deliver infants about 6 grams lighter than mothers at the 25th percentile.

In Panel B, a one unit increase in mean maternal blood lead in the second trimester lowers birthweight by 157.3 grams. Furthermore, moving a mother from the 25th to 75th percentile of blood lead (0.17 to 0.72) in the second trimester reduces birthweight by about 87 grams. Similarly, a one unit increase in mean maternal blood lead in the third trimester lowers birthweight by 75.8 grams.

Table 6. PBPK Dose-Response: Birthweight and Simulated Maternal Blood Lead (Pre-Transition)

	Birthweight
Panel A: Full pregnancy	
AUC, full pregnancy	-0.6544***
	(0.1412)
Peak blood lead	-26.5714***
	(5.7374)
Panel B: Trimester means, entered jointly	
Mean maternal blood lead, T1	-37.5363*
	(20.1494)
Mean maternal blood lead, T2	-157.2773***
	(21.4667)
Mean maternal blood lead, T3	-75.7992***
	(17.1343)
Observations	31352

Standard errors clustered by tract in parentheses. Outcome: birthweight in grams. Pre-transition TMS live births within 25 km. Panel A reports two separate regressions, one for each measure; Panel B is a single regression. All include census tract, birth-year, and birth-month fixed effects and standard maternal controls. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.6 Mechanism: Maternal-Vascular Pathway

The results in Table 2 on birthweight and gestational age are consistent with a pathway through shorter gestation rather than slower fetal growth. Lead is an established cause of elevated blood pressure in adults as it induces oxidative stress, reduces nitric oxide, and

increases vascular resistance (Navas-Acien et al., 2007). During pregnancy, elevated blood lead has been linked to gestational hypertension and preeclampsia (Poropat et al., 2018), which can reduce blood flow to the placenta. If lead from racing fuel raised maternal blood pressure, it would therefore shorten gestation and lower birthweight, which is the pattern observed in the main results.

Table 7 estimates the main specifications with gestational hypertension, eclampsia, the five-minute Apgar score, and an indicator for a low Apgar score (below 7). These outcomes are recorded consistently only on the revised birth certificate that Texas adopted in 2005. The five-minute Apgar score is not recorded before 2005, and gestational hypertension rises from 3.1-3.9 percent in 2003-04 to 5.1-5.4 percent in 2005-06, which reflects more how those conditions were reported rather than the incidence frequency. I therefore estimate the pre-transition mechanism regressions on births from 2005-06.

Before the transition, exposure over the full pregnancy raises the probability of gestational hypertension. A doubling of the lead emitted during pregnancy raises the probability by 0.98 percentage points (significant at the 1 percent level), a 19 percent increase relative to the 5.1 percent prevalence in these cohorts. Additionally, each additional race during pregnancy raises it by 0.54 percentage points (significant at the 5 percent level). The effect is mainly showing up in the third trimester. The effect of one more race in the third trimester increases the probability by 1.04 percentage points (significant at the 1 percent level), a 20 percent increase. The log emissions in the third trimester raise it by 0.18 percentage points per log point (significant at the 5 percent level). The concentration in the third trimester lines up with the clinical onset of gestational hypertension. After the transition, the full-pregnancy estimates are small and statistically insignificant.

The evidence for this channel is not consistent across specifications. Log emissions in the first trimester decrease hypertension, and the post-transition race-count estimates are negative in the second and third trimesters (-0.49 and -0.33 percentage points significant at the 10 and 5 percent levels, respectively). Since the lead hypothesis does not predict a negative association between race and hypertension after the transition, the hypertension results are therefore suggestive. They suggest a plausible biological link that is consistent with the results for gestational age and preterm birth.

The rest of the outcomes provide little additional evidence. The number of cases of eclampsia is small. One in a thousand of these births will have experienced it. The effect of one more race during pregnancy raises the probability by 0.11 percentage points (significant at the 10 percent level), and the effect is concentrated in the first trimester, although eclampsia typically develops later in pregnancy. In addition, each additional race

in the first trimester reduces the five-minute Apgar score by 0.018 point and increase the probability of a low Apgar score by 0.31 percentage-point. The full-pregnancy estimates for both Apgar outcomes are small and statistically insignificant. The effects disappear after the transition. These estimates are small, and the evidence for effects on the newborn's condition at birth is less clear than that for gestational age.

Table 7. Mechanism: Maternal-Vascular and Newborn Outcomes

Panel A: Log emissions								
	Pre-transition (leaded), births 2005 on				Post-transition (as-if-leaded, placebo)			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Full pregnancy	Hyp.	Ecl.	Apgar5	Low Apgar	Hyp.	Ecl.	Apgar5	Low Apgar
Log emissions, full pregnancy	0.0142***	0.0017	0.0009	0.0002	-0.0069	0.0001	0.0245	-0.0036
	(0.0050)	(0.0011)	(0.0144)	(0.0027)	(0.0063)	(0.0009)	(0.0308)	(0.0059)
Observations	15219	15219	14686	14686	28620	28620	27904	27904
R-squared	0.0179	0.0100	0.0122	0.0076	0.0150	0.0048	0.0133	0.0065
By trimester (T1+T2+T3 joint)								
Log emissions, T1	-0.0015**	0.0001	-0.0019	0.0004	0.0002	-0.0000	-0.0004	0.0000
	(0.0006)	(0.0001)	(0.0016)	(0.0003)	(0.0005)	(0.0001)	(0.0015)	(0.0003)
Log emissions, T2	-0.0003	-0.0000	-0.0029	0.0003	-0.0007	-0.0000	-0.0010	-0.0001
	(0.0009)	(0.0001)	(0.0018)	(0.0003)	(0.0005)	(0.0001)	(0.0016)	(0.0003)
Log emissions, T3	0.0018**	0.0001	0.0030	-0.0003	-0.0005	-0.0000	0.0009	-0.0003
	(0.0007)	(0.0002)	(0.0023)	(0.0004)	(0.0004)	(0.0001)	(0.0015)	(0.0003)
Observations	15219	15219	14686	14686	28620	28620	27904	27904
R-squared	0.0183	0.0099	0.0130	0.0080	0.0151	0.0048	0.0133	0.0065
Panel B: Cumulative race count								
	Hyp.	Ecl.	Apgar5	Low Apgar	Hyp.	Ecl.	Apgar5	Low Apgar
Full pregnancy								
Total Races, Full Pregnancy	0.0054**	0.0011*	0.0013	0.0005	-0.0011	-0.0000	0.0048	-0.0006
	(0.0024)	(0.0006)	(0.0070)	(0.0013)	(0.0016)	(0.0002)	(0.0079)	(0.0015)
Observations	15219	15219	14686	14686	28620	28620	27904	27904
R-squared	0.0177	0.0104	0.0122	0.0077	0.0150	0.0048	0.0133	0.0065
By trimester (T1+T2+T3 joint)								
Races during T1	0.0008	0.0013**	-0.0175**	0.0031**	-0.0005	-0.0001	0.0016	-0.0002
	(0.0029)	(0.0005)	(0.0086)	(0.0015)	(0.0025)	(0.0003)	(0.0094)	(0.0015)
Races during T2	0.0073*	0.0010*	-0.0171*	0.0021	-0.0049*	-0.0002	0.0011	-0.0008
	(0.0038)	(0.0006)	(0.0099)	(0.0016)	(0.0025)	(0.0003)	(0.0095)	(0.0014)
Races during T3	0.0104***	0.0009	0.0139	-0.0016	-0.0033**	0.0000	0.0060	-0.0010
	(0.0032)	(0.0007)	(0.0088)	(0.0016)	(0.0016)	(0.0003)	(0.0078)	(0.0014)
Observations	15219	15219	14686	14686	28620	28620	27904	27904
R-squared	0.0182	0.0104	0.0134	0.0084	0.0152	0.0048	0.0133	0.0065

Standard errors in parentheses. Outcomes (LPM for binary): gestational hypertension (Hyp.), eclampsia (Ecl.), Apgar 5-min score, low Apgar indicator (<7). Sample: TMS live births within 25 km. Pre-transition columns use births from 2005 on, recorded on the revised birth certificate: the Apgar score is not recorded before 2005, and recorded gestational hypertension and eclampsia change with the 2005 certificate revision. All regressions include census tract and year and month fixed effects plus standard maternal controls. Standard errors clustered by tract.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.7 Monitor-Based Exposure and Other Texas Racetracks

I construct another measure that uses EPA's AQS database for every monitor within 100 miles of TMS (Appendix I). For each race day, I take the excess lead each monitor recorded relative to its usual level, interpolate each mother's census tract, and convert it to the dose she would have absorbed by breathing. Each additional microgram of absorbed dose over the pregnancy lowers birthweight by 54.9 grams (significant at the 1 percent level), and the estimates are similar when I use only the five monitors that operated throughout the pre-transition period. Because the monitors are 44 to 57 kilometers from the track, report only once every six days, and recorded only 27 readings on pre-transition race days, I use this measure as a supporting evidence rather than the main specification.

Furthermore, there are three other NASCAR racetracks in Texas: Thunder Hill Raceway, Texas Motor Speedway, and Texas World Speedway. A natural alternative design would be to compare births near TMS to other Texas racetracks. Appendix J shows why this comparison is unreliable. Thunder Hill Raceway hosted only five regional races, whose fuel cannot be determined, and its estimates are positive both before and after the transition. Meyer Speedway in Houston and Texas World Speedway in College Station held no races during the sample period and should show no change. Texas World shows none, but birthweight near Meyer Speedway falls by about 21 grams after 2007, which likely reflects a shock specific to the Houston area. Hence, I rely on variation within TMS as a stronger alternative design.

7 Discussion and Conclusion

This paper examines the effects of prenatal lead exposure on birth outcomes by exploiting NASCAR's transition from leaded to unleaded racing fuel in 2006–2007. I find that prenatal lead exposure lowers birthweight, shortens gestation, and increases the probability of low birthweight. Each additional race during pregnancy decreases birthweight by about 14 grams, and the effects are largest for races during the second trimester. The effects are substantially weaker or absent after the transition to unleaded fuel. The reduction in birthweight is consistent with shorter gestation and a higher risk of preterm birth. The results also provide suggestive evidence that gestational hypertension may be one pathway through which lead exposure affects birth outcomes. Before the transition, mothers living closer to the racetrack and those residing downwind on race days deliver infants with lower birthweight. The distance gradient also disappears after the transition while the wind results provide additional, but less uniform, support for lead driving the effect. The PBPK model uses race timing and emissions to account for the accumulation

and persistence of lead during pregnancy. Moving from the 25th to the 75th percentile of simulated cumulative blood lead is associated with an approximately 38-gram reduction in birthweight. The EPA monitor-based construction in the Appendix also finds negative effects between lead exposure and birthweight. As lead remains present today in sources such as piston-engine aircraft and older housing paints, these results suggest that further reductions in exposure may produce health benefits for nearby communities, particularly for pregnant women and their infants.

References

ACOG (2024). How your fetus grows during pregnancy. https://www.acog.org/womens-health/faqs/how-your-fetus-grows-during-pregnancy. Accessed April 12, 2026.

Aizer, A., J. Currie, P. Simon, and P. Vivier (2018). Do low levels of blood lead reduce children's future test scores? *American Economic Journal: Applied Economics 10*(1), 307–341.

Boyle, J., D. Yeter, M. Aschner, and D. C. Wheeler (2021). Estimated iq points and lifetime earnings lost to early childhood blood lead levels in the united states. *Science of the total environment 778*, 146307.

Currie, J. (2013). Pollution and infant health. *Child development perspectives 7*(4), 237–242.

Currie, J. and M. Neidell (2005). Air pollution and infant health: what can we learn from california's recent experience? *The quarterly journal of economics 120*(3), 1003–1030.

Currie, J. and J. F. Schmieder (2008). Fetal exposure to toxic releases and infant health outcomes.

Curtis, E. M., C. Parsons, K. Maslin, S. D'angelo, R. J. Moon, S. R. Crozier, F. Gossiel, N. J. Bishop, S. H. Kennedy, A. T. Papageorghiou, et al. (2021). Bone turnover in pregnancy, measured by urinary ctx, is influenced by vitamin d supplementation and is associated with maternal bone health: findings from the maternal vitamin d osteoporosis study (mavidos) trial. *The American Journal of Clinical Nutrition 114*(5), 1600–1611.

Deryugina, T., G. Heutel, N. H. Miller, D. Molitor, and J. Reif (2019). The mortality and medical costs of air pollution: Evidence from changes in wind direction. *American Economic Review 109*(12), 4178–4219.

Hollingsworth, A., J. M. Huang, I. Rudik, and N. J. Sanders (2025). A thousand cuts: Cumulative lead exposure reduces academic achievement. *Journal of Human Resources 60*(3), 950–976.

Hollingsworth, A. and I. Rudik (2021). The effect of leaded gasoline on elderly mortality: Evidence from regulatory exemptions. *American Economic Journal: Economic Policy 13*(3), 345–373.

Jenkins, D., D. Grossman, D. Slusky, and S. Danagoulian (2022). Blood lead testing in flint before and after water contamination. *Pediatrics 150*(6), e2022056541.

Kalnay, E., M. Kanamitsu, R. Kistler, W. Collins, D. Deaven, L. Gandin, M. Iredell, S. Saha, G. White, J. Woollen, et al. (2018). The ncep/ncar 40-year reanalysis project. In *Renewable energy*, pp. Vol1_146–Vol1_194. Routledge.

Knittel, C. R., D. L. Miller, and N. J. Sanders (2016). Caution, drivers! children present: Traffic, pollution, and infant health. *Review of Economics and Statistics 98*(2), 350–366.

Lanphear, B. P., R. Hornung, J. Khoury, K. Yolton, P. Baghurst, D. C. Bellinger, R. L. Canfield, K. N. Dietrich, R. Bornschein, T. Greene, et al. (2019). Erratum:"low-level environmental lead exposure and children's intellectual function: an international pooled analysis". *Environmental health perspectives 127*(9), 099001.

Liu, J., Y. Chen, D. Gao, J. Jing, and Q. Hu (2014). Prenatal and postnatal lead exposure and cognitive development of infants followed over the first three years of life: a prospective birth study in the pearl river delta region, china. *Neurotoxicology 44*, 326–334.

Manolagas, S. C. and A. M. Parfitt (2010). What old means to bone. *Trends in Endocrinology & Metabolism 21*(6), 369–374.

Manton, W., C. Angle, K. Stanek, D. Kuntzelman, Y. Reese, and T. Kuehnemann (2003). Release of lead from bone in pregnancy and lactation. *Environmental research 92*(2), 139–151.

Navas-Acien, A., E. Guallar, E. K. Silbergeld, and S. J. Rothenberg (2007). Lead exposure and cardiovascular disease—a systematic review. *Environmental health perspectives 115*(3), 472–482.

O'Flaherty, E. J. (1991). Physiologically based models for bone-seeking elements: I. rat skeletal and bone growth. *Toxicology and applied pharmacology 111*(2), 299–312.

Oflaherty, E. J. (1993). Physiologically based models for bone-seeking elements: Iv. kinetics of lead disposition in humans. *Toxicology and applied pharmacology 118*(1), 16–29.

Oflaherty, E. J. (1995). Physiologically based models for bone-seeking elements: V. lead absorption and disposition in childhood. *Toxicology and applied pharmacology 131*(2), 297–308.

Pedersen, M., L. Giorgis-Allemand, C. Bernard, I. Aguilera, A.-M. N. Andersen, F. Ballester, R. M. Beelen, L. Chatzi, M. Cirach, A. Danileviciute, et al. (2013). Ambient air pollution and low birthweight: a european cohort study (escape). *The lancet Respiratory medicine 1*(9), 695–704.

Poropat, A. E., M. A. Laidlaw, B. Lanphear, A. Ball, and H. W. Mielke (2018). Blood lead and preeclampsia: a meta-analysis and review of implications. *Environmental research 160*, 12–19.

Rabinowitz, M. B., G. W. Wetherill, J. D. Kopple, et al. (1976). Kinetic analysis of lead metabolism in healthy humans. *The Journal of Clinical Investigation 58*(2), 260–270.

Racing-Reference (n.d.). Racing-reference: Historical racing results and statistics. https://www.racing-reference.info/.

Santana, A. B., L. E. W. Spelta, J. V. M. Sobalvarro, M. H. M. C. Podestá, R. C. T. Garcia, T. M. Dos Reis, and L. H. Torres (2023). Gestational lead exposure and its effects on fetal/infant development-a systematic review. *Reproductive Toxicology 117*, 108342.

Stieb, D. M., L. Chen, M. Eshoul, and S. Judek (2012). Ambient air pollution, birth weight and preterm birth: a systematic review and meta-analysis. *Environmental research 117*, 100–111.

Talayero, M. J., C. R. Robbins, E. R. Smith, and C. Santos-Burgoa (2023). The association between lead exposure and crime: a systematic review. *PLOS global public health 3*(8), e0002177.

Téllez-Rojo, M. M., M. Hernández-Avila, H. Lamadrid-Figueroa, D. Smith, L. Hernández-Cadena, A. Mercado, A. Aro, J. Schwartz, and H. Hu (2004). Impact of bone lead and bone resorption on plasma and whole blood lead levels during pregnancy. *American journal of epidemiology 160*(7), 668–678.

U.S. Environmental Protection Agency (2001). Review of adult lead models: Evaluation of models for assessing human health risks associated with lead exposures at non-residential areas of superfund and other hazardous waste sites. Technical report, U.S. Environmental Protection Agency, Washington, DC.

U.S. Environmental Protection Agency (2011). Exposure factors handbook 2011 edition (final report). https://cfpub.epa.gov/ncea/efp/recordisplay.cfm?deid=236252.

Yüksel, D., B. Yuksel, E. Kalafat, T. Yüce, D. C. Katlan, and A. Koç (2022). Assessment of lead and mercury levels in maternal blood, fetal cord blood and placenta in pregnancy with intrauterine growth restriction. *Journal of Basic and Clinical Health Sciences 6*(1), 199–205.

Zhu, M., E. F. Fitzgerald, K. H. Gelberg, S. Lin, and C. M. Druschel (2010). Maternal low-level lead exposure and fetal growth. *Environmental health perspectives 118*(10), 1471.

Appendix

A Balance Test

Table 8 reports the balance test referenced in Section 5.1. Each column is a separate regression of predetermined maternal characteristics on joint log emissions with census tract, birth-year, and birth-month fixed effects on the pre-transition sample. In the table, only one coefficient is statistically significant at the 10 percent level (WIC enrollment in trimester 2). A one unit increase in the log emissions in trimester 2 marginally increases WIC enrollment by 0.0024. Otherwise, all other characteristics are close to small and statistically insignificant. Therefore, mothers with more lead exposure do not differ in observable characteristics from mothers with less exposure. This suggests that race timing is unrelated to maternal composition, or in other words, mothers whose pregnancies coincided with races do not differ systematically, on observables, from those whose pregnancies did not.

Table 8. Balance Test: Log Emissions vs. Maternal Characteristics (Pre-Transition)

	(1) Mat. age	(2) Married	(3) WIC	(4) Black	(5) White	(6) First birth	(7) Insured
Log emissions, T1	0.0003	-0.0001	0.0015	-0.0004	0.0008	0.0009	-0.0003
	(0.0120)	(0.0007)	(0.0013)	(0.0004)	(0.0007)	(0.0010)	(0.0008)
Log emissions, T2	0.0057	0.0001	0.0024*	0.0004	0.0007	0.0010	0.0006
	(0.0119)	(0.0009)	(0.0014)	(0.0005)	(0.0007)	(0.0007)	(0.0009)
Log emissions, T3	-0.0016	0.0007	0.0005	0.0001	0.0003	-0.0008	0.0002
	(0.0111)	(0.0005)	(0.0009)	(0.0004)	(0.0008)	(0.0009)	(0.0009)
Observations	31362	31362	15219	31362	31362	31362	15219
R-squared	0.1339	0.0893	0.1648	0.0253	0.0336	0.0108	0.1067

Standard errors in parentheses. Each column is a separate regression of the indicated predetermined maternal characteristic on joint log emissions (T1+T2+T3) during pregnancy. Sample: TMS live births within 25 km, pre-transition (conception before Dec 17, 2005). Fixed effects: census tract, birth year and birth month. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Pregnancy Dating

Exposure windows are defined by the conception date which depends on the mother's last reported LMP. A misreported LMP can move races into the wrong window, especially for births from 2008 to 2010, and bias the estimates toward zero. Tables 9 to 11 test whether the results are sensitive to how pregnancies are dated. First, Table 9 compares LMP gestational length with the clinician's obstetric estimate of gestation, which is usually based on ultrasound. The full-pregnancy estimate is -41.6 grams per log point of emissions, compared with -38.7 grams under LMP. The race-count estimate in the second trimester is -41.0 grams, compared with -32.6 grams. The estimates become slightly larger under

the obstetric dating, although most of the effect is coming from the 2005 and later cohorts. Second, Table 10 restricts the sample to births whose two gestation measures agree exactly, differ by less than two weeks, or differ by at most two weeks. The full-pregnancy estimates range from -42.0 to -44.8 grams before the transition and are statistically insignificant after it. Third, Table 11 shifts every exposure window seven days earlier or later. The full-pregnancy estimates range from -33.1 grams and 36.5 grams and are statistically insignificant after the transition. Similarly, the trimester measures, which are the most sensitive to dating errors, also shift races within seven days of a trimester boundary. The race-count estimate in the second trimester becomes larger (-44.4 grams, compared with -32.6). Given that the results are relatively consistent with the main results in all three checks, the results therefore do not depend on how pregnancies are dated.

Table 9. Dating Conception From the Obstetric Estimate of Gestation

	Pre, all		Pre, births 2003–04		Pre, births 2005-06	
	(1) LMP	(2) Obstetric	(3) LMP	(4) Obstetric	(5) LMP	(6) Obstetric
Panel A: Log emissions, full pregnancy						
Log emissions, full pregnancy	-38.7199***	-41.5631***	6.9650	12.5668	-80.5114***	-88.9477***
	(9.6535)	(9.5203)	(17.3471)	(18.3348)	(10.8541)	(11.7777)
Observations	31353	31353	16135	16135	15218	15218
R-squared	0.0374	0.0374	0.0411	0.0412	0.0410	0.0414
Panel B: Log emissions (T1+T2+T3 joint)						
Log emissions, T1	2.1736*	2.2933*	2.1285	2.2558	2.3884*	3.0034**
	(1.2354)	(1.2079)	(2.2218)	(2.0457)	(1.3607)	(1.4766)
Log emissions, T2	0.1193	-0.2161	3.4833*	2.5345	-3.3192	-3.2030
	(1.3419)	(1.4274)	(2.0443)	(2.3814)	(2.0219)	(2.3036)
Log emissions, T3	-1.4274	-0.3721	1.7215	2.5067	-4.6763**	-3.0653
	(1.5505)	(1.5336)	(2.3608)	(2.6821)	(1.8636)	(2.0445)
Observations	31353	31353	16135	16135	15218	15218
R-squared	0.0368	0.0368	0.0414	0.0413	0.0390	0.0389
Panel C: Cumulative race count (T1+T2+T3 joint)						
Races during T1	-16.7808**	-21.7988***	6.6983	2.9602	-26.5365***	-30.8540***
	(6.3906)	(7.1942)	(13.4974)	(15.0422)	(6.7503)	(8.3195)
Races during T2	-32.6293***	-41.0404***	10.5177	2.9127	-52.7733***	-62.2532***
	(6.6665)	(8.0538)	(16.1685)	(18.6898)	(8.0472)	(9.1699)
Races during T3	-18.1562***	-16.4784***	7.8495	11.8844	-30.6827***	-29.3286***
	(5.4905)	(5.2476)	(14.7845)	(15.8942)	(6.9452)	(7.4863)
Observations	31353	31353	16135	16135	15218	15218
R-squared	0.0375	0.0379	0.0412	0.0412	0.0407	0.0415

Standard errors in parentheses. Outcome: birthweight in grams. All regressions include census tract, birth-year and birth-month fixed effects and standard maternal controls. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Odd columns date conception from the last menstrual period (birth date minus the LMP-based gestational length) as in the main tables; even columns date it from the birth date minus the clinical or obstetric estimate of gestation.

Table 10. Births Whose Two Gestation Measures Agree

	Pre-transition				Post-transition (placebo)			
	(1) All	(2) Exact	(3) <2 wks	(4) ≤2 wks	(5) All	(6) Exact	(7) <2 wks	(8) ≤2 wks
Panel A: Log emissions, full pregnancy								
Log emissions, full pregnancy	-38.7199***	-42.9330***	-44.7604***	-41.9536***	0.1794	26.6427	4.7564	-0.3846
	(9.6535)	(14.4190)	(10.5718)	(10.4141)	(19.6126)	(31.1663)	(27.1000)	(25.0719)
Observations	31353	17160	26034	28655	28620	17492	24000	26360
R-squared	0.0374	0.0375	0.0369	0.0366	0.0405	0.0415	0.0406	0.0410
Panel B: Log emissions (T1+T2+T3 joint)								
Log emissions, T1	2.1736*	2.1423	1.9179	2.0178	-2.0111	-0.7516	-2.5298	-2.7104*
	(1.2354)	(1.7759)	(1.5094)	(1.2487)	(1.2790)	(1.8926)	(1.7441)	(1.6109)
Log emissions, T2	0.1193	0.9604	1.0350	0.6472	-0.6628	2.1354	1.5282	0.6902
	(1.3419)	(1.8244)	(1.6864)	(1.4088)	(1.4632)	(2.4660)	(1.8078)	(1.6643)
Log emissions, T3	-1.4274	0.0788	-0.5995	-0.6460	0.5444	2.5643	1.3238	1.1992
	(1.5505)	(2.2540)	(1.7088)	(1.7237)	(1.3006)	(1.8488)	(1.5378)	(1.4563)
Observations	31353	17160	26034	28655	28620	17492	24000	26360
R-squared	0.0368	0.0368	0.0361	0.0359	0.0406	0.0416	0.0409	0.0413
Panel C: Cumulative race count (T1+T2+T3 joint)								
Races during T1	-16.7808**	-24.1951**	-23.6787***	-19.2809***	-7.7685	-0.7648	-4.9102	-7.7765
	(6.3906)	(9.7399)	(7.3056)	(6.7576)	(6.4732)	(10.1636)	(8.5276)	(8.0346)
Races during T2	-32.6293***	-40.0456***	-37.3320***	-34.6179***	-6.2113	2.3612	4.7234	-1.1733
	(6.6665)	(9.5227)	(7.6615)	(7.1417)	(7.0154)	(13.5502)	(9.5704)	(8.6875)
Races during T3	-18.1562***	-15.1254**	-17.2682***	-16.2932***	1.6833	7.9556	5.4838	3.1267
	(5.4905)	(6.8134)	(5.4520)	(5.6042)	(5.4976)	(8.7275)	(7.2283)	(6.6498)
Observations	31353	17160	26034	28655	28620	17492	24000	26360
R-squared	0.0375	0.0378	0.0370	0.0367	0.0406	0.0415	0.0407	0.0411

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km. Pre-transition: conceived before 17 December 2005; post-transition: conceived on or after 1 March 2007, where log emissions are as-if-leaded (placebo). Samples are defined by the LMP-based conception date in every column. All regressions include census tract, birth-year and birth-month fixed effects and standard maternal controls. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns keep all births, or births whose gestational length from the last menstrual period and the clinical or obstetric estimate of gestation agree exactly, differ by less than two weeks, or differ by at most two weeks (both measures are in whole weeks). Conception dates, and hence trimester windows, are based on the last menstrual period.

Table 11. Shifting and Buffering the Trimester Windows

	Pre-transition				Post-transition (placebo)			
	(1) Main	(2) −7 days	(3) +7 days	(4) Buffer	(5) Main	(6) −7 days	(7) +7 days	(8) Buffer
Panel A: Log emissions, full pregnancy								
Log emissions, full pregnancy	-38.7199***	-36.5248***	-33.1211***	-38.7199***	0.1794	2.8416	6.9532	0.1794
	(9.6535)	(11.7518)	(7.9334)	(9.6535)	(19.6126)	(15.6083)	(21.2951)	(19.6126)
Observations	31353	31353	31353	31353	28620	28620	28620	28620
R-squared	0.0374	0.0373	0.0372	0.0374	0.0405	0.0405	0.0405	0.0405
Panel B: Log emissions (T1+T2+T3 joint)								
Log emissions, T1	2.1736*	2.1708	2.5219**	0.8885	-2.0111	-0.4235	-0.7382	-0.9817
	(1.2354)	(1.4254)	(1.1346)	(1.1255)	(1.2790)	(1.2860)	(1.4349)	(1.2858)
Log emissions, T2	0.1193	-1.4324	-0.6662	-3.2944*	-0.6628	1.5129	1.7318	2.4976*
	(1.3419)	(1.7415)	(1.3153)	(1.6496)	(1.4632)	(1.4235)	(1.0768)	(1.4451)
Log emissions, T3	-1.4274	-2.1564	-0.5964	-1.8759	0.5444	1.1696	1.4305	1.8451
	(1.5505)	(1.5613)	(1.3335)	(1.5616)	(1.3006)	(1.1222)	(1.2282)	(1.2530)
Observations	31353	31353	31353	31353	28620	28620	28620	28620
R-squared	0.0368	0.0370	0.0369	0.0370	0.0406	0.0406	0.0407	0.0407
Panel C: Cumulative race count (T1+T2+T3 joint)								
Races during T1	-16.7808**	-15.9862*	-15.2317***	-19.6651***	-7.7685	0.9610	-1.8294	-6.1273
	(6.3906)	(8.2306)	(4.8796)	(7.0217)	(6.4732)	(6.1833)	(7.2219)	(7.0879)
Races during T2	-32.6293***	-37.3878***	-32.6468***	-44.3725***	-6.2113	-1.2253	2.2354	-2.6558
	(6.6665)	(6.9264)	(6.4504)	(7.7350)	(7.0154)	(7.0102)	(6.2305)	(8.3167)
Races during T3	-18.1562***	-18.6186***	-15.4285***	-18.9400***	1.6833	-2.7867	2.8618	0.8799
	(5.4905)	(5.1684)	(5.1640)	(5.2340)	(5.4976)	(5.5921)	(4.7199)	(5.3072)
Observations	31353	31353	31353	31353	28620	28620	28620	28620
R-squared	0.0375	0.0378	0.0375	0.0380	0.0406	0.0405	0.0405	0.0406

Standard errors in parentheses. All regressions include census tract, birth-year and birth-month fixed effects and standard maternal controls. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Main: trimester windows from the LMP-based conception date, as in the main tables. ±7 days: every window shifted 7 days earlier or later. Buffer: races within 7 days of the first/second or second/third trimester cutoff are removed from the trimester exposure measures.

C Functional Form

The main specification uses the log plus one transformation. Table 12 examines the sensitivity of the results to alternative transformations. Across all transformations except in levels (Column 5), the pre-transition estimates are similar for the full pregnancy while the post-transition estimates are close to zero in every column. However, the trimester estimates are inconsistent pre-transition, which is another reason why I use race counts for the trimester results.

Table 12. Transformation of Lead Emissions

	(1) Log(1 + E), grams	(2) Log E	(3) Inverse hyperbolic sine	(4) Log(1 + E), kilograms	(5) E, kilograms
Panel A: Pre-transition, full-pregnancy exposure					
Birthweight	-38.7199***	-38.7173***	-38.7173***	-41.2270***	-1.3141***
	(9.6535)	(9.6529)	(9.6529)	(10.2099)	(0.3076)
Gestational age	-0.2064***	-0.2063***	-0.2063***	-0.2190***	-0.0065***
	(0.0471)	(0.0471)	(0.0471)	(0.0498)	(0.0015)
Low birthweight	0.0106***	0.0106***	0.0106***	0.0113***	0.0003**
	(0.0039)	(0.0039)	(0.0039)	(0.0042)	(0.0001)
Panel B: Post-transition, full-pregnancy exposure					
Birthweight	0.1794	0.1794	0.1794	0.1962	0.0103
	(19.6126)	(19.6123)	(19.6123)	(19.9772)	(0.3502)
Gestational age	0.0877	0.0877	0.0877	0.0894	0.0016
	(0.0823)	(0.0823)	(0.0823)	(0.0838)	(0.0015)
Low birthweight	0.0003	0.0003	0.0003	0.0003	-0.0000
	(0.0095)	(0.0095)	(0.0095)	(0.0096)	(0.0002)
Panel C: Pre-transition, trimester exposure (birthweight)					
Trimester 1	2.1736*	–	2.0778*	-0.6881	-1.7482***
	(1.2354)		(1.1497)	(4.3406)	(0.4688)
Trimester 2	0.1193	–	0.1925	-10.6301**	-2.8342***
	(1.3419)		(1.2518)	(4.3448)	(0.4660)
Trimester 3	-1.4274	–	-1.2974	-8.8820*	-1.2708***
	(1.5505)		(1.4502)	(4.5256)	(0.3416)
Panel D: Post-transition, trimester exposure (birthweight)					
Trimester 1	-2.0111	–	-1.8941	-4.9069	-0.2467
	(1.2790)		(1.1933)	(4.3712)	(0.5049)
Trimester 2	-0.6628	–	-0.6067	-3.1096	-0.2256
	(1.4632)		(1.3630)	(5.1451)	(0.5218)
Trimester 3	0.5444	–	0.5136	1.0759	0.1561
	(1.3006)		(1.2143)	(4.2617)	(0.4072)

Standard errors clustered by tract in parentheses. Controls and fixed effects as in Table 2. Panels A and B: birthweight (grams), gestational age (weeks), and low birthweight; Panels C and D: birthweight, with the three trimesters entered jointly. Column 1 is the main specification; column 2 is undefined for trimesters with no race. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13 examines whether the linear specification correctly captures the relationship. In Panel A, which adds the square of full pregnancy log emissions, the squared term is negative and significant in the full sample but is not significant for births from 2005-06 before the transition even though the magnitude is the same. It is insignificant for births in 2003-04 and after the transition. Panel B adds the square of emissions in kilograms instead. The squared term is positive and significant only for births from 2005-06. The squared terms are significant only in some samples, and their sign depends on whether emissions are measured in logs or in kilograms. The slopes in both panels are constant ranging from -56.2 to 61.7 grams per log point and -1.39 to -1.5 grams per log kilogram. Hence, the relationship implied by the quadratics is close to linear.

As discussed in Section 5, the trimester log-emission estimates in Table 2 are near zero because about a third of mothers have no race in a given trimester, and therefore, a single slope must fit both the jump from zero to about 9 log points and the small differences among exposed mothers. Panel C separates these two parts. Each trimester includes an

indicator for whether the mother had any race, and for mothers who did, how much lead those races emitted. Prior to the transition, for mothers with at least one race, more lead lowers birthweight in each trimester (-44.9, -62.5, and -22.5 grams per log point) whereas the estimates after the transition are small. The trimester log-emission estimates in the main table are close to zero because one slope cannot fit both parts, not because lead has no effect in a given trimester. For this reason, I use the number of races in each trimester for the trimester results.

Splitting trimester exposure this way does not change the cohort difference. Among exposed mothers, the estimates are large for births from 2005-06 and close to zero for births in 2003–04, as are the full-pregnancy slopes in Panels A and B. The cohort difference in Table 4 is therefore not a result of the linear functional form. Panel D adds squared terms to trimester emissions in kilograms. Most of the estimates are insignificant before the transition and after it.

Table 13. Functional Form of the Dose-Response

	Pre-transition			Post-transition
	(1) All	(2) Births 2003–04	(3) Births 2005-06	(4) (placebo)
Panel A: Quadratic in log emissions, full pregnancy				
Log emissions, full pregnancy	280.6357*	-236.9660	239.8789	-972.0583
	(153.2168)	(546.2914)	(199.3647)	(2299.1967)
Log emissions, full pregnancy, squared	-15.9864**	12.4418	-15.9866	44.5753
	(7.7411)	(28.1283)	(10.0607)	(105.3465)
Slope at 25th percentile	-56.17	25.16	-96.93	-32.94
	(13.85)	(49.40)	(16.60)	(82.02)
Slope at median	-58.45	26.93	-99.21	-26.58
	(14.67)	(53.19)	(17.73)	(67.54)
Slope at 75th percentile	-61.70	29.47	-102.46	-17.50
	(15.92)	(58.67)	(19.42)	(47.41)
p-value, squared term	0.043	0.660	0.117	0.674
Observations	31353	16135	15218	28620
Panel B: Quadratic in emissions (kilograms), full pregnancy				
Emissions (kg), full pregnancy	-2.0496***	0.6475	-5.1424***	-0.0859
	(0.7029)	(3.4076)	(0.7955)	(4.2470)
Emissions (kg), full pregnancy, squared	0.0074	-0.0048	0.0280***	0.0008
	(0.0058)	(0.0659)	(0.0070)	(0.0362)
Slope at 25th percentile	-1.49	0.29	-3.04	-0.02
	(0.36)	(1.86)	(0.41)	(1.55)
Slope at median	-1.45	0.26	-2.88	-0.02
	(0.35)	(2.20)	(0.39)	(1.35)
Slope at 75th percentile	-1.39	0.22	-2.64	-0.01
	(0.32)	(2.73)	(0.37)	(1.05)
p-value, squared term	0.210	0.942	0.000	0.982
Observations	31353	16135	15218	28620
Panel C: Trimester exposure split into any race and emissions among exposed				
Any race, T1	15.7544	12.3636	44.7903***	-49.5188***
	(11.2913)	(21.2671)	(13.2971)	(16.9747)
Any race, T2	-67.5322***	40.5007	-120.6694***	-10.1004
	(13.8980)	(35.6223)	(25.0001)	(18.8223)
Any race, T3	0.4778	22.8573	-11.7451	-1.1013
	(14.6368)	(21.0932)	(19.2558)	(13.4982)
Log emissions among exposed, T1	-44.8844***	3.0938	-77.7387***	15.7996
	(10.0961)	(21.0242)	(10.9229)	(11.9216)
Log emissions among exposed, T2	-62.4974***	-2.5260	-102.9687***	-17.3320*
	(10.1929)	(20.4813)	(11.3165)	(10.1580)
Log emissions among exposed, T3	-22.5198**	21.1195	-44.5912**	-3.4485
	(8.5531)	(20.4489)	(16.9943)	(11.6245)
Observations	31353	16135	15218	28620
Panel D: Quadratic in emissions (kilograms), by trimester				
Emissions (kg), T1	1.3202	1.7780	3.0508	-1.7737
	(1.6392)	(2.7735)	(2.0206)	(1.8003)
Emissions (kg), T2	0.0704	6.1878*	-3.2406	1.7207
	(1.5827)	(3.3304)	(3.1735)	(1.7920)
Emissions (kg), T3	-0.8777	0.3933	-2.9027	-2.7148
	(1.8159)	(3.1399)	(2.3395)	(1.9942)
Emissions (kg), T1, squared	-0.0763**	-0.0331	-0.1331***	0.0408
	(0.0381)	(0.0740)	(0.0449)	(0.0481)
Emissions (kg), T2, squared	-0.0677*	-0.1402*	-0.0255	-0.0462
	(0.0395)	(0.0785)	(0.0697)	(0.0412)
Emissions (kg), T3, squared	-0.0039	0.0261	0.0173	0.0778
	(0.0413)	(0.0923)	(0.0586)	(0.0503)
p-value, squared terms (joint)	0.153	0.198	0.032	0.382
Observations	31353	16135	15218	28620

Standard errors in parentheses. All regressions include census tract, birth-year and birth-month fixed effects and standard maternal controls. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panels A and B add the square of full-pregnancy log emissions and of emissions in kilograms, respectively. Slopes are evaluated at the 25th percentile, median, and 75th percentile of pre-transition exposure. Panel C replaces each trimester's log emissions with an indicator for any race and, for exposed mothers, log emissions relative to the pre-transition exposed median. Panel D adds squares of each trimester's emissions in kilograms.

D Sources of Exposure Variation

Table 14 shows where the exposure variation comes from once the census tract, birth-year, and birth-month fixed effects are included as discussed in Section 5.1. There are two sources of variation: 1) changes in race calendar across years and 2) timing of conception among mothers delivering in the same month and year. These sources of variation could overlap with other changes such as local economic shock or a unusually hot season that affect births in a particular month and year. In addition, a race close to either the start or the end of the 280-day pregnancy might be counted for some mothers and not for others. Panel A splits the variation in race counts and log emissions left after the fixed effects into variation explained by race calendar of each year and among births in the same month and year. Before the transition, about a third of the variation comes from the race calendar and two-thirds from births in the same month and year. After the transition, almost all of the variation comes from births in the same month and year when the race schedule did not change (no second race weekend after 2005). In Panel B, about half comes from races in the last 45 days of the window and about half comes from the last 45 days of the window. In short, most of the variation comes from births born just before or after a race and conceived just before or after a race. Panel C splits the variation in log emissions into the number of races, mix of Cup, Xfinity, and truck races, and differences in the lead emitted by the same type of race across years. The number of races explained more than 99 percent of the variation before the transition so the race-count and log-emission measures rely on the same variation. Panel D replaces the birth-year and birth-month fixed effects with birth year-by-month fixed effects which removes the variation from the race calendar and leaves only the variation among births in the same month and year. The pre-transition log-emission estimate falls in magnitude from -38.7 to -30.3 grams and remains significant at the 5 percent level. The race-count estimate falls in magnitude from -14.5 to -9.1 grams and is no longer statistically significant. The post-transition estimates remain statistically insignificant. The pre-transition effect therefore does not depend on changes in the race calendar across years.

Table 14. Sources of Exposure Variation

	Pre-transition		Post-transition	
	Races	Log emissions	Races	Log emissions
Panel A: Variation left after the fixed effects				
Standard deviation	1.256	0.624	1.264	0.309
Standard deviation after fixed effects	0.752	0.366	0.782	0.194
Share from the race calendar of each year	0.359	0.335	0.024	0.064
Share among births in the same month and year	0.641	0.665	0.976	0.936
Panel B: Race counts among births in the same month and year				
Share of the variation from races in				
The last 45 days of the window (born just before or after a race)	0.483		0.502	
The first 45 days of the window (conceived just before or after a race)	0.503		0.514	
Days 45–235 of the window	0.014		-0.016	
Share of mothers whose race count differs from the most common count among births in the same month and year	0.182		0.173	
Panel C: Log emissions, share of the variation left after the fixed effects				
Explained by the number of races		0.997		0.968
Explained additionally by the mix of Cup, Xfinity, and truck races		0.001		0.007
From differences in lead emitted by the same type of race across years		0.002		0.025
Panel D: Birthweight estimates				
Birth-year and birth-month fixed effects (main)	-14.4901***	-38.7199***	-1.8793	0.1794
	(4.6666)	(9.6535)	(4.5715)	(19.6126)
Birth year-by-month fixed effects	-9.1127	-30.2940**	2.5763	18.7726
	(6.9436)	(14.4485)	(4.5525)	(19.6082)
Observations	31352	31352	28619	28619

Exposure is measured over the full pregnancy. Panel A splits the variance left after the tract, birth-year, and birth-month fixed effects into the part absorbed by birth year-by-month fixed effects and the rest. Panel C: series are inferred from the weekend schedule. Panel D: outcome is birthweight in grams; standard errors clustered by tract in parentheses; controls as in Table 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Cohort Difference Before the Transition

Table 15 examines why the pre-transition estimates are close to zero for births in 2003-04 and large for births from 2005-06 (Table 4). It tests for 1) birth year-by-month fixed effects, 2) no fall weekend race during pregnancy, 3) controls recorded in both cohorts only, 4) LMP and clinical gestion agree, and 5) conception dated from obstetric estimate. For births from 2003-04, the estimates are similar except the last two tests where the estimates increase to 22.3 and 12.6, respectively, but are still statistically insignificant. For births from 2005, the estimates are similar and statitically significant across all tests. None of these tests accounts for the differences, and I cannot explain it with the available data.

Table 15. Pre-Transition Cohort Difference: Births 2003-04 and 2005-06

	Births 2003-04	Births 2005-06
SD of log emissions after fixed effects	0.327	0.351
(1) Main specification	6.9650	-80.5114***
	(17.3471)	(10.8541)
(2) Birth year-by-month fixed effects	6.6743	-90.8553***
	(17.5112)	(16.2145)
(3) No fall-weekend race during pregnancy	–	-96.3159***
		(20.4447)
(4) Controls recorded in both cohorts only	6.9650	-79.8606***
	(17.3471)	(10.8371)
(5) LMP and clinical gestation agree	22.3271	-95.5513***
	(30.2322)	(16.5094)
(6) Conception dated from obstetric estimate	12.5668	-88.9477***
	(18.3348)	(11.7777)
Observations (main)	16135	15218

Outcome: birthweight in grams. Standard errors clustered by tract in parentheses. Pre-transition births only; controls and fixed effects as in Table 2 unless noted. Row 3 is not defined for 2003–04, when there was no fall Cup weekend. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Exposure Timing and Gestation Length

The birthweight and gestational age results are consistent with a pathway through shorter gestation. Since both exposure and gestational length are measured using conception date, this section checks whether this relationship is an effect of how exposure is defined. Because the exposure window runs from conception date to full term of 280 days regardless of when the baby was born, it counts races that occurred after the birth for preterm births. Furthermore, the regressions hold the birth month fixed and conception is dated by counting back from the birth date. Hence, for two births on the same day, the shorter pregnancy has a later window, and exposure could be mechanically related to gestation length. So exposure could differ between shorter and longer pregnancies purely because of how the windows are defined. Table 16 tests both.

Columns 2 and 3 end the window at day 259 (37 weeks) or day 195 (the end of the second trimester) so races after a term birth or after almost any birth are not counted. The pre-transition estimates are similar (-32.8 and -35.8 grams per log point compared with -38.7). I do not end each window at the actual birth date because that would give shorter pregnancies fewer races by construction. Exposure is constructed the same way after the transition when the estimates are close to zero. This is reassuring but it does not rule out a mechanical relationship because the sources of variation differ across the two periods (Appendix D).

Columns 4 and 5 address the concern that shorter pregnancies could imply a later conception date more directly by replacing birth-year and birth-month fixed effects with con-

ception year-by-month fixed effects so that mothers are compared only with others who conceived in the same month. This removes the pooled pre-transition effect, although the estimate for births from 2005-06 remains negative and significant (-44.6 grams, not shown). These fixed effects absorb all variation in conception months and leave only variation in conception date within a month. Looking at Column 5 with the 195 day window, the pre-transition results are sensitive to how conception cohorts are controlled for, and I cannot explain yet why. Finally, I restrict the same to term births to examine how much of the effect remains when we do not shorten gestation (Column 6). The estimate falls to -11.8 grams. Since this restriction conditions on gestation, which is an outcome itself, I interpret the result as a descriptive comparison and not a mechanism test. In the post-transition sample, two race count estimates (Columns 2 and 5) are negative and significant at the 10 percent level.

Table 16. Exposure Timing and Gestation Length

	(1) Main	(2) Day 259	(3) Day 195	(4) Conception FE	(5) Day 195 + conc. FE	(6) Term only
Panel A: Pre-transition, log emissions						
Log emissions in window (among exposed)	-38.7199***	-32.8095***	-35.7906***	-2.2381	26.3227**	-11.8058*
	(9.6535)	(8.5779)	(8.2222)	(14.2557)	(10.9808)	(6.1530)
Observations	31353	31353	31353	31353	31353	28763
Panel B: Pre-transition, number of races						
Races in window	-14.4901***	-15.2381***	-16.5009***	0.6774	13.5563**	-4.5708
	(4.6666)	(4.5764)	(5.5117)	(6.7379)	(6.1702)	(2.9368)
Observations	31353	31353	31353	31353	31353	28763
Panel C: Post-transition (placebo), log emissions						
Log emissions in window (among exposed)	0.1794	-26.1134	1.5754	31.3848	-1.0760	-5.8837
	(19.6126)	(18.9109)	(11.1987)	(29.5809)	(12.8089)	(14.1195)
Observations	28620	28620	28620	28620	28620	26299
Panel D: Post-transition (placebo), number of races						
Races in window	-1.8793	-8.3826*	-7.8433	4.9974	-11.6774*	-1.2114
	(4.5715)	(4.4686)	(4.9660)	(7.7971)	(6.3719)	(3.5252)
Observations	28620	28620	28620	28620	28620	26299

Standard errors in parentheses. Outcome: birthweight in grams. All regressions include census tract, birth-year and birth-month fixed effects and standard maternal controls. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns: (1) main specification; (2) and (3) windows end at day 259 (37 weeks) or day 195 (end of the second trimester); (4) conception year-by-month fixed effects replace birth-year and birth-month fixed effects; (5) columns 3 and 4 combined; (6) births at 37-41 weeks only. Exposure is divided into an indicator for whether a race occurred during the window (not shown) and log emissions relative to the pre-transition median for mothers exposed to at least one race. The reported coefficient captures the effect of higher lead emissions among these mothers.

Table 17 shows how much variation is absorbed by the conception year-by-month fixed effects. Before the transition, the standard deviation of full-pregnancy log emissions falls from 0.37 to 0.22 after replacing birth-year and birth-month fixed effects with conception year-by-month fixed effects. After controlling for conception year-by-month fixed effects, about a third of the baseline variation is left (34 percent for log emissions and 29 percent

for race counts). Most of the variation absorbed by these fixed effects comes from differences in conception timing relative to the race calendar. Gestational length accounts for less than 1 percent of the baseline variation. The remaining variation comes from mothers who conceived in the same month but on different days. Panel B shows that roughly 60 percent of the variation comes from races in the first 45 days of the window and 40 percent from races in the last 45 days before the transition. It is difficult to tell whether these fixed effects remove meaningful variation or confounding by conception cohort.

Table 17. Exposure Variation Under Conception Year-by-Month Fixed Effects

	Pre-transition		Post-transition	
	Races	Log emissions	Races	Log emissions
Panel A: Variation left after the fixed effects				
Standard deviation, birth-year and birth-month fixed effects (main)	0.752	0.366	0.782	0.194
Standard deviation, conception year-by-month fixed effects	0.414	0.217	0.478	0.120
Share of the main variation left when conception year-month is also held fixed	0.293	0.341	0.357	0.369
Share of the main variation explained by gestational length (LMP, weeks)	0.003	0.005	0.008	0.008
Panel B: Race counts among births conceived in the same month and year				
Share of the variation from races in				
The last 45 days of the window	0.424		0.291	
The first 45 days of the window	0.604		0.640	
Days 45–235 of the window	-0.029		0.069	

All rows use full-pregnancy exposure. The main fixed effects are census tract, birth year, and birth month, as in the main tables. The conception fixed effects are census tract and conception year-by-month, as in column 4 of Table 16. Row 3 adds conception year-by-month to the main fixed effects and reports the share of the variance that remains.Row 4 reports the within R-squared from regressing exposure on LMP-based gestational-week indicators, controlling for the main fixed effects.Panel B shows how the remaining variation in race counts is distributed across different parts of the window after controlling for conception fixed effects. The shares sum to one.

G Gestation Outcomes

Table 18 examines how the effect varies across the gestational age. Before the transition, a one-log-point increase in emissions during pregnancy raises the probability of preterm birth (less than 37 weeks) by 2.4 percentage points and the probability of early preterm birth (less than 34 weeks) by 1.0 percentage point, both significant at the 1 percent level, relative to prevalences of 7.8 and 1.6 percent. The effect on early-term birth (37 to 38 weeks) is smaller and marginally significant. Each additional race during pregnancy raises the probability of preterm birth by 0.9 percentage points. After the transition, all estimates are small and statistically insignificant. The reduction in gestation is therefore concentrated among preterm births, which is consistent with the gestational hypertension channel in Section 6.6.

Table 18. Exposure and the Distribution of Gestational Age

	Preterm (<37 wks)		Early preterm (<34 wks)		Early term (37–38 wks)	
	(1) Pre	(2) Post	(3) Pre	(4) Post	(5) Pre	(6) Post
Panel A: Log emissions, full pregnancy						
Log emissions, full pregnancy	0.0236***	-0.0098	0.0098***	-0.0036	0.0093*	-0.0096
	(0.0071)	(0.0105)	(0.0033)	(0.0068)	(0.0052)	(0.0126)
Mean of outcome	0.078	0.077	0.016	0.017	0.355	0.336
Observations	31362	28620	31362	28620	31362	28620
Panel B: Number of races, full pregnancy						
Total races, full pregnancy	0.0090***	-0.0017	0.0038**	0.0001	0.0011	-0.0039
	(0.0033)	(0.0025)	(0.0014)	(0.0017)	(0.0027)	(0.0029)
Mean of outcome	0.078	0.077	0.016	0.017	0.355	0.336
Observations	31362	28620	31362	28620	31362	28620

Standard errors in parentheses. Linear probability models. Outcomes are indicators based on the clinical or obstetric estimate of gestation: preterm (less than 37 weeks), early preterm (less than 34 weeks), and early term (37–38 weeks). All regressions include census tract, birth-year and birth-month fixed effects and standard maternal controls. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H Wind Direction: Seasonality

With census tract fixed effects, downwind exposure varies depending on which races occur during a mother's pregnancy. Since wind direction varies by season, downwind status may capture seasonal patterns that birth-month fixed effects do not fully account for. Table 19 tests this using placebo days, when there were no race emissions: downwind status on the same weekday 7 or 14 days before each race (Columns 2 and 3) and the share of all pregnancy days on which the mother was downwind (Column 4). Before the transition, placebo downwind status is unrelated to birthweight, suggesting that the effect is specific to race days. When the placebo measures are controlled for (Column 5), the pre-transition estimate becomes larger (-52.7 grams) while the post-transition estimate falls from 32.7 to -11.8 grams and is statistically insignificant. Adding census tract by conception-quarter fixed effects (Column 6) or controlling for log emissions instead of the number of races (Column 7) leaves the pre-transition estimate negative and significant.

Table 19. Wind Direction: Placebo Days and Seasonality

	(1) Race day	(2) 7 days before	(3) 14 days before	(4) All days	(5) Joint	(6) Tract × season FE	(7) Emissions control
Panel A: Pre-transition							
Downwind share, race days	-39.8192**				-52.7306***	-99.9679***	-35.4960**
	(15.1708)				(15.3565)	(35.8092)	(14.5718)
Downwind share, 7 days before		-7.7362			-19.6565		
		(13.7105)			(16.6424)		
Downwind share, 14 days before			-6.2609		-11.1791		
			(14.4780)		(15.7901)		
Downwind share, all days				274.2551**	374.6974***		
				(129.5499)	(136.5218)		
Observations	31353	31353	31353	31353	31353	31335	31353
Panel B: Post-transition (placebo)							
Downwind share, race days	32.7107*				-11.7853	79.8384	32.6606*
	(18.9357)				(26.9409)	(56.6997)	(19.0137)
Downwind share, 7 days before		20.5464			41.1066*		
		(13.1193)			(22.4102)		
Downwind share, 14 days before			-10.2560		-13.2023		
			(17.0158)		(17.4520)		
Downwind share, all days				-344.9245***	-419.0623***		
				(77.3573)	(108.4317)		
Observations	28620	28620	28620	28620	28620	28595	28620

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km. Pre-transition: conceived before 17 December 2005; post-transition: conceived on or after 1 March 2007. All regressions include census tract, birth-year and birth-month fixed effects, standard maternal controls and the number of races during pregnancy (log emissions in column 7). Standard errors clustered by tract. A mother is downwind on a day if the wind blows toward her residence within 45 degrees of the bearing from the track. Columns: (1) share of race days during pregnancy on which the mother was downwind, as in Table 5; (2) and (3) the same races, with downwind status taken from the wind 7 or 14 days before each race (same weekday, no race emissions); (4) share of all days in the pregnancy window on which the mother was downwind (within-tract standard deviation 0.025 before and 0.028 after the transition); (5) the race-day share and the three placebo measures together; (6) adds census tract by conception-quarter fixed effects; (7) controls for full-pregnancy log emissions instead of the number of races. Daily wind is from the NCEP/NCAR reanalysis at the grid point nearest TMS, the same source as the race-day wind. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I Monitor-Based Exposure

The EPA collects air-quality readings from monitor stations. These stations report lead concentration in the air every 6 days. During the sample period, there are seven monitors that operate within 100 miles of TMS. Because they report only every 6 days, there are 27 readings for race days before the transition, and about a half of them comes from a single monitor. I use the excess lead a monitor recorded on a race day, relative to its usual level, to approximate the extra lead in the air that day. Then I interpolate it to each mother's home and convert it to the dose she would have absorbed by breathing. This measure is noisier than the emissions-based measure because the monitors are far from the track and readings on race days are sparse.

For each race day, I compute an inverse distance weighted (IDW) ambient lead concen-

tration at each mother's residence using the EPA lead monitors within 100 miles (about 161 km) of TMS.

$$w_{im} = \frac{1}{\max(d_{im}, 1)}, \qquad \text{Lead}_{ij} = \frac{\sum_m w_{im} \cdot \text{ExcessLead}_{mj}}{\sum_m w_{im}} \tag{21}$$

where w_{im} is the inverse-distance weight assigned to monitor m for mother i; d_{im} is the distance in kilometers between mother i's census tract centroid and monitor m (monitors beyond 150 km receive zero weight); ExcessLead_{mj} is the race-day excess lead concentration at monitor m on race day j (the monitor's reading minus its annual non-race-day baseline); and Lead_{ij} is the IDW ambient lead concentration assigned to mother i on race day j measured in $\mu\text{g/m}^3$.

The race-day absorbed dose for mother i on race day j is then:

$$q_{ij}^A = a \cdot \text{Lead}_{ij} \cdot IR_i \cdot \Delta t_{ij} \tag{22}$$

where q_{ij}^A is the absorbed lead dose (measured in micrograms) for mother i on race day j; Lead_{ij} is the IDW ambient lead concentration from Equation 21; $a = 0.3$ is the standard pulmonary absorption fraction (Oflaherty, 1993); $IR_i = 16$ m^3/day is the maternal inhalation rate (U.S. Environmental Protection Agency, 2011); and $\Delta t = 1$ day represents the 24-hour averaging period of the monitor readings. For mothers without monitor coverage during a given trimester, the assigned exposure is set to zero. Because lead readily crosses the placenta and lead levels in maternal and umbilical cord blood are highly correlated, I use maternal blood lead as a proxy for fetal lead exposure.

Table 20 lists the EPA monitors used in this construction and their race-day coverage before and after the transition.

Table 20. EPA AQS lead monitors at TMS: race-day coverage 2003–2010

Site ID	County	Distance	Years active	Total readings	Race days (pre)	Race days (post)	Mean excess Pb	Consistent
480850009	Collin	43.7	2003–2010	469	3	3	0.000	✓
480850003	Collin	44.0	2003–2010	463	3	3	0.000	✓
480850007	Collin	44.0	2003–2010	463	3	3	0.000	✓
481130057	Dallas	47.8	2003–2007	1393	14	0	0.023	✓
481130018	Dallas	56.9	2003–2007	239	3	0	0.024	✓
481130061	Dallas	47.1	2003–2003	8	0	0	–	
481130066	Dallas	57.2	2003–2004	101	1	0	0.046	

Notes: These EPA Air Quality System (AQS) ambient lead monitors are within 100 miles of TMS with at least one valid reading during 2003-2010. Distance is the distance from the monitor's reported coordinates to TMS. "Years active" is the year in which the monitor reported any reading, regardless of whether the reading fell on a race day. "Total readings" is the count of all valid daily readings the monitor reported over 2003-2010.

"Race days (pre)" counts the number of TMS race days on which this monitor reported a reading restricted to race days through Dec. 17, 2005; "Race days (post)" covers race days from Mar. 1, 2007 onward. "Mean excess Pb" is the monitor's mean race-day reading minus its annual non-race-day baseline, averaged over its pre-transition race day (in $\mu g/m^3$); "-" indicates the monitor recorded no pre-transition race-day readings.

"Consistent" comprises monitors that operated continuously throughout the pre-transition window (2003-2006) and is used in the robustness check below. Site 481130066 is excluded because it dropped out after 2004. Site 481130061 is excluded because it operated only briefly in 2003 and caught no race days. Of the 17 pre-transition race-day readings in the consistent set, 14 come from site 481130057 in Dallas County.

Table 21 reports the results in three panels. In Panel A, an extra microgram of absorbed dose above the pregnancy reduces birthweight by 54.9 grams (significant at the 1 percent level). The first and second trimesters have the largest effects (-84.0 and -72.8 grams). Panel B shows the coefficients when we pass the absorbed dose into the PBPK model to simulate blood lead levels. Both cumulative and peak blood lead lower birthweight. The first and second trimesters have the largest effects. Panel C reports the race counts for the same births. Both the monitor- and emission-based measures indicate that higher prenatal lead exposure lowers birthweight.

Table 21. Model A Calibration: Monitor-Based Exposure (Pre-Transition)

	Birthweight
Panel A: Absorbed dose (monitor-based)	
Full pregnancy	
Absorbed dose, full pregnancy	-54.8990***
	(10.8432)
Trimesters, entered jointly	
Absorbed dose, T1	-83.9527***
	(18.4791)
Absorbed dose, T2	-72.7578***
	(15.4137)
Absorbed dose, T3	-29.0710*
	(16.3186)
Observations	31353
Panel B: PBPK simulated maternal blood lead	
Full pregnancy	
AUC, full pregnancy	-2.0742***
	(0.3808)
Peak blood lead	-85.8640***
	(16.2297)
Trimesters, entered jointly	
Mean maternal blood lead, T1	-265.2753***
	(70.1192)
Mean maternal blood lead, T2	-330.3412***
	(64.2617)
Mean maternal blood lead, T3	-79.5750
	(52.4508)
Observations	31352
Panel C: Race count	
Full pregnancy	
Races, full pregnancy	-14.4901***
	(4.6666)
Trimesters, entered jointly	
Races during T1	-16.7808**
	(6.3906)
Races during T2	-32.6293***
	(6.6665)
Races during T3	-18.1562***
	(5.4905)
Observations	31353

Standard errors clustered by tract in parentheses. Outcome: birthweight in grams. Pre-transition TMS live births within 25 km. Each full-pregnancy row is a separate regression, and the trimester rows in each panel are one regression. All include census tract, birth-year, and birth-month fixed effects and standard maternal controls. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A potential concern with using EPA monitors is that EPA monitor coverage near TMS changed over the sample period. Of the seven monitors used in the monitor-based specification, two dropped out during the pre-transition window (site 481130066 after 2004 and site 481130061 after 2003). This creates a potential issue of monitor attrition. To address this, I re-estimate the model's regressions using only the five monitors active throughout the entire pre-transition window (2003–2006) and compare the results side by side. Table 22 reports the birthweight in grams.

In Panel A, the first-trimester coefficient on absorbed dose is -84.0 grams with all seven monitors and -84.9 grams with the five consistent monitors, and the second-trimester coefficients are -72.8 and -73.4 grams, all significant at the 1 percent level. The third-trimester coefficients are smaller (-29.1 and -28.0 grams) and significant only at the 10 percent level. In Panel B, the PBPK blood lead coefficients are -265.3 and -268.2 grams in the first trimester and -330.3 and -333.8 grams in the second, all significant at the 1 percent level, while the third-trimester coefficients are not significant. In Panel C, the estimates are almost the same (-2.07), both significant at the 1 percent level. The standard errors are similar across specifications. Since the estimates are very close from both monitor sets, this suggests that the results are not sensitive to excluding the two monitors that stopped reporting earlier.

Table 22. Consistent-Monitor Robustness

	(1) Full (7 monitors)	(2) Consistent (5 monitors)
Panel A: Absorbed dose (joint, pre-transition)		
Absorbed dose, T1	-83.9527***	-84.8786***
	(18.4791)	(18.4547)
Absorbed dose, T2	-72.7578***	-73.4291***
	(15.4137)	(15.4086)
Absorbed dose, T3	-29.0710*	-27.9721*
	(16.3186)	(16.3922)
Observations	31353	31353
R-squared	0.0376	0.0376
Panel B: PBPK trimester means (joint, pre-transition)		
	Full (7 monitors)	Consistent (5 monitors)
Mean maternal blood lead, T1	-265.2753***	-268.2469***
	(70.1192)	(69.4412)
Mean maternal blood lead, T2	-330.3412***	-333.8280***
	(64.2617)	(64.4282)
Mean maternal blood lead, T3	-79.5750	-76.4436
	(52.4508)	(52.6014)
Observations	31352	31352
R-squared	0.0378	0.0379
Panel C: Cumulative dose (pre-transition)		
	Full (7 monitors)	Consistent (5 monitors)
Cumulative dose (AUC)	-2.0742***	-2.0765***
	(0.3808)	(0.3794)
Observations	31352	31352
R-squared	0.0377	0.0377

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km, pre-transition only. Fixed effects: census tract, year, month. Standard errors clustered by tract. Full (7 monitors) uses 7 EPA monitors within 100 miles of TMS active during 2003-2010. Consistent (5 monitors) restricts to the 5 monitors active throughout 2003-2006. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J Threats to Cross-Track Comparisons

NASCAR also held races at venues other than TMS. Thunder Hill Raceway hosted five regional races between 2005 and 2009, while Meyer Speedway and Texas World Speedway are legacy racetracks with no NASCAR activity during the sample period. Table 24 examines whether using these venues in a cross-track comparison would affect identification

relative to the within-TMS design. Panel A reports the results for Thunder Hill Raceway. Each additional race during the second trimester is associated with birthweight that is 58 grams higher before the transition and 57 grams higher after it. The post-transition estimates are positive and significant in every trimester. Thunder Hill hosted only five regional races in which lead status cannot be determined, making it an unreliable comparison site for identifying the effects of lead exposure. In Panel B, Meyer Speedway exhibits a negative and statistically significant post-transition coefficient of approximately 21 grams, whereas Texas World Speedway shows no significant change. The decline in birthweight at Meyer Speedway may reflect a shock in the Houston area after 2007 rather than NASCAR-related factors. It could potentially be driven by local demographic and economic changes around the 2008 financial crisis. Given that there is no similar decline at Texas World Speedway, the change may be specific to the Houston area rather than reflecting a general effect of lead exposure. These results highlight a potential problem with using births near Meyer as a control group for those near TMS. The within-TMS design avoids this issue by restricting the sample to mothers living near TMS and exploiting variation in race exposure within census tracts and across birth years and months. This approach addresses threats to cross-track comparisons that may be affected by different regional trends.

Table 23 summarizes the characteristics of the four Texas racetracks.

Table 23. NASCAR Racetracks in Texas

Track Name	City	Lap Length	Period Active	Role
Texas Motor Speedway	Fort Worth	1.5 mi	1996–present	**Primary**
Thunder Hill Raceway	Kyle	0.375 mi	1998–2009	Excluded
Texas World Speedway	College Station	2.0 mi	1969–1993	Legacy track
Meyer Speedway	Houston	0.5 mi	1969–1971	Legacy track

Notes: Thunder Hill Raceway is excluded because the leaded-fuel status of its five regional NASCAR races between 2005 and 2009 cannot be determined.

Table 24. Replication and Falsification at Other Texas Tracks

Panel A: Thunder Hill — log emissions (pre — post)		
	(1) Pre-transition	(2) Post-transition (as-if-leaded, placebo)
Log emissions, T1	5.1876	10.5619***
	(4.0912)	(1.5096)
Log emissions, T2	8.2568***	8.0944**
	(1.4533)	(2.9912)
Log emissions, T3	-1.4376	7.5511*
	(4.3610)	(3.6924)
Observations	7990	9023
R-squared	0.0595	0.0450
Log emissions, full pregnancy	3.2223	9.0405***
	(2.3935)	(1.7777)
Observations	7990	9023
R-squared	0.0591	0.0449
Panel A (cont.): Thunder Hill — cumulative race count (pre + post)		
	Pre-transition	Post-transition (placebo)
Races during T1	36.6862	74.2926***
	(28.9324)	(10.6586)
Races during T2	58.3911***	56.6937**
	(10.2774)	(21.4817)
Races during T3	-10.1668	53.4222*
	(30.8404)	(25.8872)
Observations	7990	9023
R-squared	0.0595	0.0450
Total Races, Full Pregnancy	22.7875	63.5335***
	(16.9264)	(12.9839)
Observations	7990	9023
R-squared	0.0591	0.0449
Panel B: Post-2007 birthweight break at legacy tracks (falsification)		
	Meyer (Houston)	TX World (College Stn.)
Post-transition	-20.6948***	1.7425
	(4.6906)	(16.5292)
Year trend	2.6481**	-3.6785
	(1.3212)	(5.2537)
Observations	284224	15532
R-squared	0.0491	0.0512

Standard errors in parentheses. Outcome: birthweight in grams. City fixed effects due to the small geographic area. Sample: births within 25 km. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

K Race Schedule

Table 25 lists every NASCAR race at the four Texas tracks during the sample period (2003-2010). TMS hosted multiple national-series races each year, typically clustering around two weekends in April and November. Thunder Hill hosted a small number of KNPSW regional races between 2005 and 2009. Following the national switch, the races at Thunder Hill from March 31, 2007 would have unleaded while the race on October 1, 2005 would be leaded. The status for the race on September 30, 2006 is undetermined. Texas World Speedway and Meyer Speedway hosted no races during the sample period (last active in 1993 and 1971, respectively) and are omitted from this table.

Table 25. NASCAR Race Schedule at Texas Tracks, 2003–2010

Track	Date	Series	Fuel	Miles	Drivers
Thunder Hill	10/01/2005	KNPSW	Leaded	1132	20
Thunder Hill	09/30/2006	KNPSW	Leaded	1156	22
Thunder Hill	03/31/2007	KNPSW	Unleaded	1472	28
Thunder Hill	04/19/2008	KNPSW	Unleaded	1188	22
Thunder Hill	03/14/2009	KNPSW	Unleaded	974	19
TMS	03/29/2003	Xfinity	Leaded	10236	43
TMS	03/30/2003	Cup	Leaded	19646	43
TMS	06/06/2003	CWTS	Leaded	6538	36
TMS	10/11/2003	CWTS	Leaded	6537	36
TMS	04/03/2004	Xfinity	Leaded	10576	43
TMS	04/04/2004	Cup	Leaded	19016	43
TMS	06/11/2004	CWTS	Leaded	6651	34
TMS	10/16/2004	CWTS	Leaded	6568	36
TMS	04/16/2005	Xfinity	Leaded	11724	43
TMS	04/17/2005	Cup	Leaded	19006	43
TMS	06/10/2005	CWTS	Leaded	8067	36
TMS	11/04/2005	CWTS	Leaded	7262	36
TMS	11/05/2005	Xfinity	Leaded	11922	43
TMS	11/06/2005	Cup	Leaded	20862	43
TMS	04/08/2006	Xfinity	Leaded	12855	43
TMS	04/09/2006	Cup	Leaded	19456	43
TMS	06/09/2006	CWTS	Leaded	8409	36
TMS	11/03/2006	CWTS	Unleaded	7240	36
TMS	11/04/2006	Xfinity	Unleaded	11392	43
TMS	11/05/2006	Cup	Leaded	20432	43
TMS	04/14/2007	Xfinity	Unleaded	10738	43
TMS	04/15/2007	Cup	Unleaded	19338	43
TMS	06/08/2007	CWTS	Unleaded	6768	35
TMS	11/02/2007	CWTS	Unleaded	6540	36
TMS	11/03/2007	Xfinity	Unleaded	11133	43
TMS	11/04/2007	Cup	Unleaded	19131	43
TMS	04/05/2008	Xfinity	Unleaded	10794	43
TMS	04/06/2008	Cup	Unleaded	20853	43
TMS	06/06/2008	CWTS	Unleaded	7508	35
TMS	10/31/2008	CWTS	Unleaded	5216	34
TMS	11/01/2008	Xfinity	Unleaded	10635	43
TMS	11/02/2008	Cup	Unleaded	21098	43
TMS	04/04/2009	Xfinity	Unleaded	11266	43
TMS	04/05/2009	Cup	Unleaded	19972	43
TMS	06/05/2009	CWTS	Unleaded	5848	33
TMS	11/06/2009	CWTS	Unleaded	5968	36
TMS	11/07/2009	Xfinity	Unleaded	8836	43
TMS	11/08/2009	Cup	Unleaded	18784	43
TMS	04/16/2010	Arca	Unleaded	4350	38
TMS	04/19/2010	Cup	Unleaded	19040	43
TMS	04/19/2010	Xfinity	Unleaded	10138	43
TMS	06/04/2010	CWTS	Unleaded	5962	36
TMS	11/05/2010	CWTS	Unleaded	6591	36
TMS	11/06/2010	Xfinity	Unleaded	10760	43
TMS	11/07/2010	Cup	Unleaded	18435	43

TMS hosted Cup, Xfinity, and Camping World Truck Series in most years, typically in Spring and Fall. Thunder Hill hosted five KNPSW regional races between 2005 and 2009. Texas World Speedway and Meyer Speedway hosted no races during the sample period (last active in 1993 and 1971, respectively) and are omitted from this table. Fuel indicates whether the race used leaded or unleaded gasoline.